Inverse Trigonometric Functions

DISCLAIMER

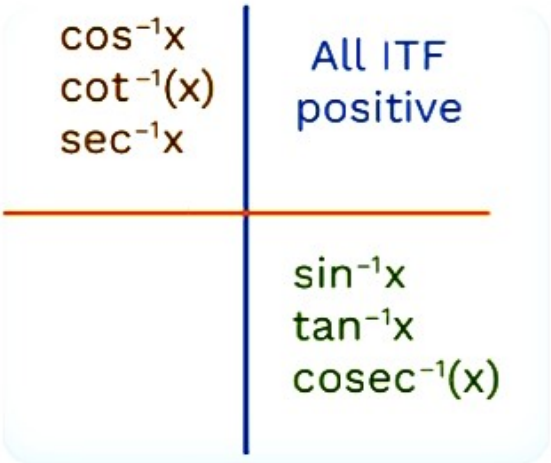
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Inverse Trigonometric Functions

All trigonometric functions are not invertible. To make them invertible we cut their domain.

$\sin^{-1}x$ denote the angle or real number whose sine is x. Similarly, $\cos^{-1}x$ and $\tan^{-1}x$ denote the angle or real number whose cosine and tangent is x, respectively.

ITF	DOMAIN	RANGE
$\sin^{-1}x$	$[-1, 1]$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
$\cos^{-1}x$	$[-1, 1]$	$[0, \pi]$
$\tan^{-1}x$	R	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
$\text{cosec}^{-1}x$	$(-\infty, -1] \cup [1, \infty)$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$
$\sec^{-1}x$	$(-\infty, -1] \cup [1, \infty)$	$[0, \pi] - \left\{\frac{\pi}{2}\right\}$
$\cot^{-1}x$	R	$(0, \pi)$



Point to Remember!!!

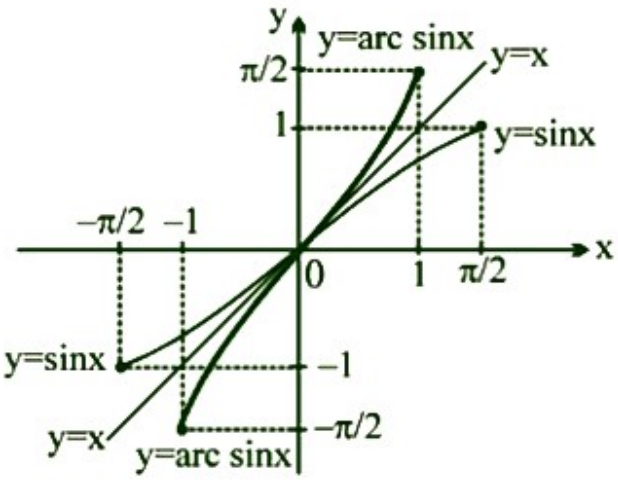
(i) 1st quadrant common to all ITF.
(ii) 3rd quadrant is not used in ITF.
(iii) 4th quadrant is not used in the clockwise direction.

Point to Remember!!!

1. All inverse trigonometric functions are bounded.
2. Any inverse trigonometric function is reflection of respective trigonometric function w.r.t. y = x line.

Graph of $\sin^{-1}x$:

1. Maximum value is $\frac{\pi}{2}$ and minimum value is $-\frac{\pi}{2}$.
2. If $y = \sin^{-1}x$ then $\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$.

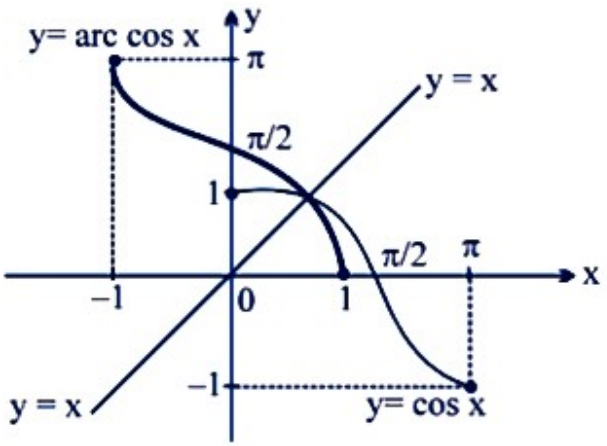


Properties of $\sin^{-1}x$ Graph :

1. Graph is aperiodic.
2. $\sin^{-1}x$ is bounded.
3. $\sin^{-1}x$ is odd function.
4. $\sin^{-1}x$ is an increasing function.

Graph of $\cos^{-1}x$:

1. Maximum value is π and minimum value is 0.
2. If $y = \cos^{-1}x$ then $\frac{dy}{dx} = -\frac{1}{\sqrt{1-x^2}}$.

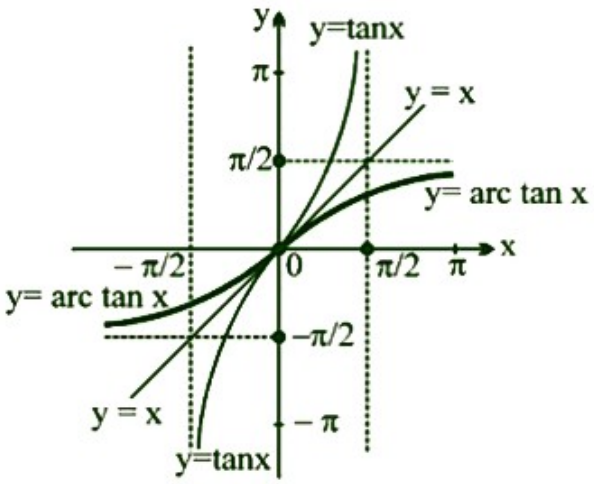


Properties of $\cos^{-1}x$ Graph :

1. Graph is aperiodic.
2. $\cos^{-1}x$ is bounded.
3. $\cos^{-1}x$ is neither even nor odd function.
4. $\cos^{-1}x$ is a decreasing function.

Graph of $\tan^{-1}x$:

1. Maximum value and minimum value doesn't exist.
2. If $y = \tan^{-1}x$ then $\frac{dy}{dx} = \frac{1}{1+x^2}$.



Properties of $\tan^{-1}x$ Graph :

1. Graph is aperiodic.
2. $\tan^{-1}x$ is bounded.
3. $\tan^{-1}x$ is odd function.
4. $\tan^{-1}x$ is an increasing function.

Know the facts

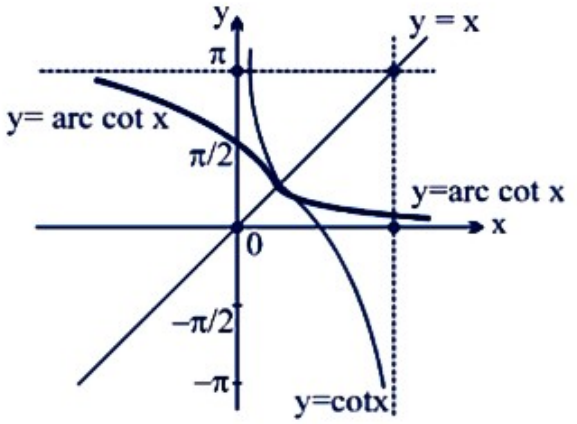
$\tan^{-1}x > \tan^{-1}y$

$\Rightarrow x > y$

(as function is increasing)

Graph of $\cot^{-1}x$:

1. Maximum value and minimum value doesn't exist.
2. If $y = \cot^{-1}x$ then $\frac{dy}{dx} = -\frac{1}{1+x^2}$.



Properties of $\cot^{-1}x$ Graph :

1. Graph is aperiodic.
2. $\cot^{-1}x$ is bounded.
3. $\cot^{-1}x$ is neither even nor odd function.
4. $\cot^{-1}x$ is a decreasing function.

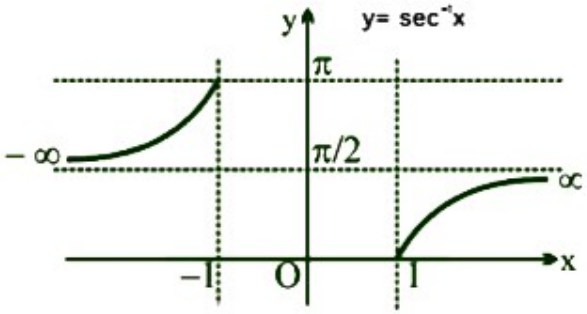
Know the facts

$\cot^{-1}x > \cot^{-1}y$

$\Rightarrow x < y$

(as function is decreasing)

Graph of $\sec^{-1}x$:

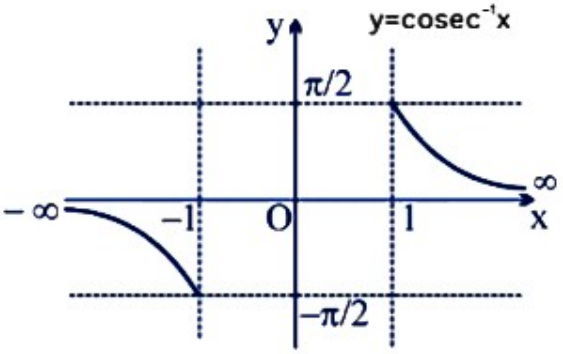


1. Maximum value is π and minimum value is 0.
2. If $y = \sec^{-1}x$ then $\frac{dy}{dx} = \frac{1}{|x|\sqrt{x^2-1}}$.

Properties of $\sec^{-1}x$ Graph :

1. Graph is aperiodic.
2. $\sec^{-1}x$ is bounded.
3. $\sec^{-1}x$ is neither even nor odd function.
4. $\sec^{-1}x$ is an increasing in both of its branches.

Graph of $\text{cosec}^{-1}x$:



1. Maximum value is $\frac{\pi}{2}$ and minimum value is $-\frac{\pi}{2}$.
2. If $y = \text{cosec}^{-1}x$ then $\frac{dy}{dx} = -\frac{1}{|x|\sqrt{x^2-1}}$.

Properties of $\text{cosec}^{-1}x$ Graph :

1. Graph is aperiodic.
2. $\text{cosec}^{-1}x$ is bounded.
3. $\text{cosec}^{-1}x$ is odd function.
4. $\text{cosec}^{-1}x$ is a decreasing in both of its branches.

Note :

If a function is continuous and monotonic on R ⇏ Range of function is R.

e.g. $\tan^{-1}x$ and $\cot^{-1}x$

If a function is continuous and has range R ⇏ function is monotonic.

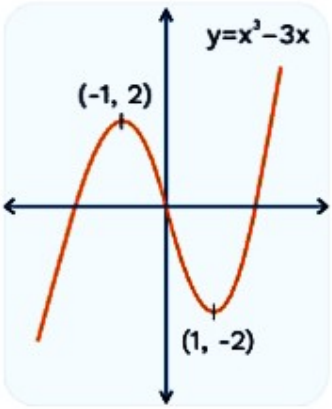
e.g. $y = x^3 - 3x$

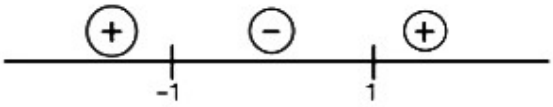
$\frac{dy}{dx} = y' = 3x^2 - 3 = 3(x-1)(x+1)$

Sign of y'

y is increasing in $(-\infty, -1) \cup (1, \infty)$

y is decreasing in $(-1, 1)$





ILLUSTRATIONS :

Q1 **Tell domain and range of $y = \sin^{-1}(e^x)$.**

A1 Domain: $-1 \le e^x \le 1$

$\Rightarrow e^x \le e^0$ (As if we take real power of positive number it is always positive, so e^x is always greater than -1)

$\Rightarrow x \le 0$

Range: As $x \le 0$

$\Rightarrow x \in (-\infty, 0]$

$\Rightarrow e^x \in (e^{-\infty}, e^0]$

$\Rightarrow e^x \in (0, 1]$

$\Rightarrow \sin^{-1}(e^x) \in (\sin^{-1}0, \sin^{-1}1]$

$\Rightarrow \sin^{-1}(e^x) \in \left(0, \frac{\pi}{2}\right]$

Domain: $(-\infty, 0]$; Range: $\left(0, \frac{\pi}{2}\right]$

Q2 **Tell domain and range of $y = \cos^{-1}[x]$, where [.] denotes greatest integer function.**

A2 Domain: $-1 \le [x] \le 1$

$-1 \le x < 2$

Range: As $-1 \le x < 2$

$\Rightarrow x \in [-1, 2)$

$\Rightarrow x \in [-1, 0) \cup [0, 1) \cup [1, 2)$

$\Rightarrow [x] \in \{-1, 0, 1\}$

$\Rightarrow \cos^{-1}[x] \in \{\cos^{-1}(-1), \cos^{-1}(0), \cos^{-1}(1)\}$

$\Rightarrow \cos^{-1}[x] \in \left\{\pi, \frac{\pi}{2}, 0\right\}$

Domain: [−1. 2); Range: $\left\{\pi, \frac{\pi}{2}, 0\right\}$

Q3 **Tell domain and range of $y = \sin^{-1}\{x\}$, where {.} denotes fractional part function.**

A3 Domain: $-1 \le \{x\} \le 1$

$\Rightarrow x \in R$

Range: As $x \in R$

$\Rightarrow \{x\} \in [0, 1)$

$\Rightarrow \sin^{-1}\{x\} \in [\sin^{-1}(0), \sin^{-1}(1))$ (As $\sin^{-1}x$ is increasing function)

$\Rightarrow \sin^{-1}\{x\} \in \left[0, \frac{\pi}{2}\right)$

Domain: R; Range: $\left[0, \frac{\pi}{2}\right)$

Q4 **Tell domain and range of $y = \cos^{-1}\{x\}$, where {.} denotes fractional part function.**

A4 Domain: $-1 \le \{x\} \le 1$

$\Rightarrow x \in R$

Range: As $x \in R$

$\Rightarrow \{x\} \in [0, 1)$

$\Rightarrow \cos^{-1}\{x\} \in (\cos^{-1}(1), \cos^{-1}(0)]$ (As $\cos^{-1}x$ is decreasing function)

$\Rightarrow \cos^{-1}\{x\} \in \left(0, \frac{\pi}{2}\right]$

Domain: R; Range: $\left(0, \frac{\pi}{2}\right]$

Q5 **Tell domain and range of $y = \cot^{-1}(\text{sgn } x)$.**

A5 Domain: As sgn(x) is well defined for $x \in R$ and gives {−1, 0, 1} as output where $\cot^{-1}x$ is well defined

$\Rightarrow x \in R$

Range: As $x \in R$

$\Rightarrow \text{sgn } x \in \{-1, 0, 1\}$

$\Rightarrow \cot^{-1}(\text{sgn } x) \in \{\cot^{-1}(-1), \cot^{-1}(0), \cot^{-1}(1)\}$

$\Rightarrow \cot^{-1}(\text{sgn } x) \in \left\{\frac{3\pi}{4}, \frac{\pi}{2}, \frac{\pi}{4}\right\}$

Domain: R; Range: $\left\{\frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}\right\}$

Q6 **Tell domain and range of $y = \tan^{-1}(\log_2(x^2 - 2x + 2))$**

A6 Domain:

i) $x^2 - 2x + 2 > 0$ (because of log)

$\Rightarrow (x - 1)^2 + 1 > 0 \Rightarrow x \in R$

ii) $\log_2(x^2 - 2x + 2) \in R$ (because of $\tan^{-1}$)

$\Rightarrow x \in R$

So, Domain is R

Range:

$x^2 - 2x + 2 = (x - 1)^2 + 1$

$(x - 1)^2 + 1 \geq 1$

$\log_2 (x^2 - 2x + 2) \geq \log_2 1$ (As graph of $\log_2 x$ is increasing)

$\Rightarrow \log_2 (x^2 - 2x + 2) \geq 0$

$\Rightarrow \tan^{-1}(\log_2 (x^2 - 2x + 2)) \geq \tan^{-1}(0)$ (As graph of $\tan^{-1}x$ is increasing)

$\Rightarrow y \geq 0$

Domain: R; Range: $\left[0, \frac{\pi}{2}\right)$

Q7 **Tell the value of** $\sin^{-1}\left(-\frac{1}{\sqrt{2}}\right) + \cos^{-1}\left(-\frac{1}{\sqrt{2}}\right) - \tan^{-1}\left(-\sqrt{3}\right) + \cot^{-1}\left(-\frac{1}{\sqrt{3}}\right)$

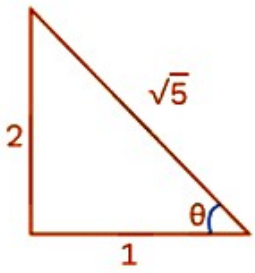
A7 $-\frac{\pi}{4} + \frac{3\pi}{4} - \left(-\frac{\pi}{3}\right) + \frac{2\pi}{3} = \frac{3\pi}{2}$

Q8 **Tell the value of $\sin(\tan^{-1}2)$**

A8 Let $\tan^{-1} 2 = \theta$

$\Rightarrow 2 = \tan\theta$

$\sin\left(\tan^{-1} 2\right) = \sin\theta = \frac{2}{\sqrt{5}}$

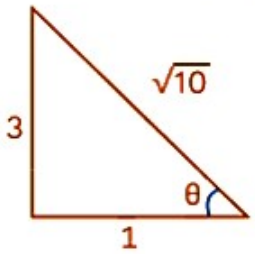


Q9 **Tell the value of $\cos(\tan^{-1}3)$**

A9 Let $\tan^{-1} 3 = \theta$

$\Rightarrow 3 = \tan\theta$

$$\cos\left(\tan^{-1} 3\right) = \cos(\theta) = \frac{1}{\sqrt{10}}$$



Q10 **Tell the value of $\sin\left(2\sin^{-1}\left(\frac{3}{5}\right)\right)$**

A10 let $\sin^{-1}\left(\frac{3}{5}\right) = \theta$ (note that θ lies in first quadrant)

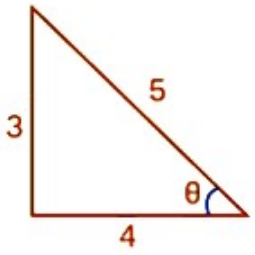
$$\Rightarrow \frac{3}{5} = \sin\theta$$

$$\sin\left(2\sin^{-1}\left(\frac{3}{5}\right)\right) = \sin 2\theta$$

$$= 2\sin\theta\cos\theta$$

$$= 2\left(\frac{3}{5}\right)\left(\frac{4}{5}\right)$$

$$= \frac{24}{25}$$



Q11 **Tell the value of $\cos(2\tan^{-1}2)$**

A11 Let $\tan^{-1} 2 = \theta$

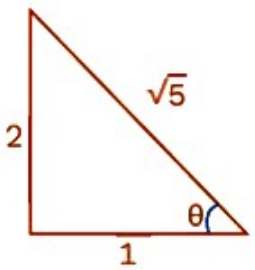
$\Rightarrow 2 = \tan\theta$

$$\cos\left(2\tan^{-1} 2\right) = \cos(2\theta)$$

$$= \frac{1-\tan^2\theta}{1+\tan^2\theta}$$

$$= \frac{1-4}{1+4}$$

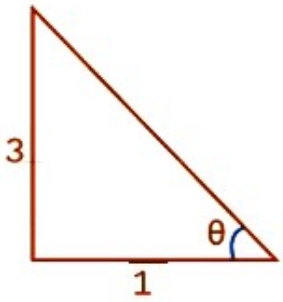
$$= -\frac{3}{5}$$



Q12 **Tell the value of $\cos(2\tan^{-1}3)$**

A12 Let $\tan^{-1}3 = \theta$

$\Rightarrow 3 = \tan\theta$



$$\cos\left(2\tan^{-1}3\right) = \cos 2\theta$$

$$= \frac{1-\tan^2\theta}{1+\tan^2\theta}$$

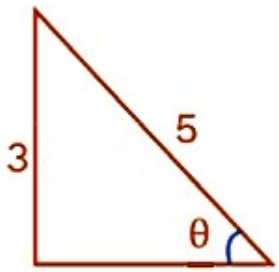
$$= \frac{1-9}{1+9}$$

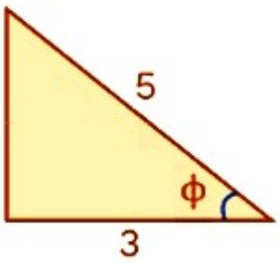
$$= -\frac{4}{5}$$

Q13 **Tell the value of** $\sin\left(\arcsin\left(\frac{3}{5}\right) - \arccos\left(\frac{3}{5}\right)\right)$

A13 Let $\sin^{-1}\left(\frac{3}{5}\right) = \theta$ and $\cos^{-1}\left(\frac{3}{5}\right) = \phi$

$\Rightarrow \left(\frac{3}{5}\right) = \sin\theta$ and $\left(\frac{3}{5}\right) = \cos\phi$





$$\sin\left(\sin^{-1}\frac{3}{5} - \cos^{-1}\frac{3}{5}\right) = \sin(\theta - \phi)$$

$$= \sin\theta\cos\phi - \cos\theta\sin\phi$$

$$= \left(\frac{3}{5}\right)\left(\frac{3}{5}\right) - \left(\frac{4}{5}\right)\left(\frac{4}{5}\right)$$

$$= -\frac{7}{25}$$

Q14 **Tell the value of** $\sin\left(\tan^{-1}\left(\cos\left(\cot^{-1}\left(-\frac{1}{\sqrt{3}}\right)\right)\right)\right)$

A14

$$\cot^{-1}\left(-\frac{1}{\sqrt{3}}\right)=\frac{2\pi}{3}$$

$$\cos\left(\cot^{-1}\left(-\frac{1}{\sqrt{3}}\right)\right)=\cos\frac{2\pi}{3}$$

$$=-\frac{1}{2}$$

$$\tan^{-1}\left(\cos\left(\cot^{-1}\left(-\frac{1}{\sqrt{3}}\right)\right)\right)=\tan^{-1}\left(-\frac{1}{2}\right)$$

$$=-\tan^{-1}\left(\frac{1}{2}\right)$$

$$\sin\left(\tan^{-1}\left(\cos\left(\cot^{-1}\left(-\frac{1}{\sqrt{3}}\right)\right)\right)\right)=\sin\left(-\tan^{-1}\left(\frac{1}{2}\right)\right)$$

$$=-\sin\left(\tan^{-1}\left(\frac{1}{2}\right)\right)$$

$$=-\frac{1}{\sqrt{5}}$$

Q15 **Show that** $\cos\left(\tan^{-1}\sin\cot^{-1}x\right)=\sqrt{\left(\frac{x^2+1}{x^2+2}\right)}, x\in(0,1)$

A15 Let $\cot^{-1}x=\theta$ (note that $\theta\in(0,\pi)$)

$$x=\cot\theta$$

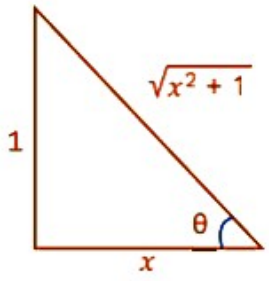
$$\sin\left(\cot^{-1}x\right)=\sin\theta$$

$$=\frac{1}{\sqrt{x^2+1}}$$

$$\tan^{-1}\left(\sin\left(\cot^{-1}x\right)\right)=\tan^{-1}\left(\frac{1}{\sqrt{x^2+1}}\right)$$

$$\cos\left(\tan^{-1}\left(\sin\left(\cot^{-1}x\right)\right)\right)=\cos\left(\tan^{-1}\left(\frac{1}{\sqrt{x^2+1}}\right)\right)$$

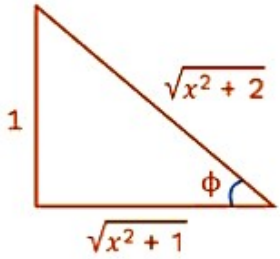
$$=\cos\phi$$



where, $\phi = \tan^{-1}\left(\frac{1}{\sqrt{x^2+1}}\right) \Rightarrow \tan\phi = \frac{1}{\sqrt{x^2+1}}$

$\cos\left(\tan^{-1}\left(\sin\left(\cot^{-1}x\right)\right)\right) = \cos\phi$

$= \sqrt{\frac{x^2+1}{x^2+2}}$



Q16 **If $\cos^{-1}x + \cos^{-1}y = 0$, then find the value of $x + y$.**

A16 Minimum of L.H.S. $= 0 + 0$ (Range of $\cos^{-1}x$ is $[0, \pi]$)

$= 0$

$= \text{R.H.S.}$

$\Rightarrow \cos^{-1}x = \cos^{-1}y = 0$

$\Rightarrow x = y = \cos 0$

$\Rightarrow x = y = 1$

$\Rightarrow x + y = 2$

Q17 **If $\cos^{-1}x + \cos^{-1}y + \cos^{-1}z = 3\pi$, then find the value of $x + y + z$.**

A17 Maximum of L.H.S. $= \pi + \pi + \pi$ (Range of $\cos^{-1}x$ is $[0, \pi]$)

$= 3\pi$

$= \text{R.H.S.}$

$\Rightarrow \cos^{-1}x = \cos^{-1}y = \cos^{-1}z = \pi$

$\Rightarrow x = y = z = \cos\pi$

$\Rightarrow x = y = z = -1$

$\Rightarrow x + y + z = -3$

Q18 **If $\sin^{-1}x + \sin^{-1}y + \sin^{-1}z = \frac{3\pi}{2}$, then find the value of $x + y + z$.**

A18 Maximum value of L.H.S. $= \frac{\pi}{2} + \frac{\pi}{2} + \frac{\pi}{2}$ (Range of $\sin^{-1}x$ is $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$)

$= \frac{3\pi}{2} = \text{R.H.S.}$

$\Rightarrow \sin^{-1}x = \sin^{-1}y = \sin^{-1}z = \frac{\pi}{2} \Rightarrow x = y = z = \sin\frac{\pi}{2}$

$\Rightarrow x = y = z = 1$

$\Rightarrow x + y + z = 3$

Q19 **True/False :**

f(x) = sgn (arc cot x) and g(x) = $\sin^2 x + \cos^2 x$ are identical functions.

A19 Let $f(x) = \text{sgn}(\cot^{-1} x)$

Domain: R,

Now, $f(x) = \text{sgn}(0, \pi)$ (As range of $\cot^{-1}x$ in $(0, \pi)$)

$f(x) = 1$

Again Let $g(x) = \sin^2 x + \cos^2 x$

Domain: R, $g(x) = 1$

$\Rightarrow$ f(x) and g(x) are identical functions.

So, statement is true.

Q20 **If f:R → $\left[0, \frac{\pi}{2}\right)$, f(x) = $\tan^{-1}(x^2 + 2x + \alpha)$ is onto function then find α.**

A20 As, function is onto:

Co-Domain = Range

$\left[0, \frac{\pi}{2}\right)$ = Range

So, $\tan^{-1}\left(x^2 + 2x + \alpha\right) \in \left[0, \frac{\pi}{2}\right)$

$\Rightarrow$ $x^2 + 2x + \alpha \in \left[\tan 0, \tan\frac{\pi}{2}\right)$

$\Rightarrow$ $x^2 + 2x + \alpha \in [0, \infty)$

$\Rightarrow$ Range of $x^2 + 2x + \alpha$ is $[0, \infty)$

$\Rightarrow$ $-\frac{D}{4a} = 0$ (Range of quadratic (a>0) is $\left[-\frac{D}{4a}, \infty\right)$)

$\Rightarrow$ $-\frac{(4 - 4\alpha)}{4(1)} = 0$

$\Rightarrow$ $\alpha = 1$

Q21 **If $f : R \to \left[-\frac{\pi}{4}, \frac{\pi}{2}\right)$, $f(x) = \tan^{-1}(x^2 + 2x + \alpha)$ is onto function then find α.**

A21 As, function is onto:
Co-Domain = Range

$\left[-\frac{\pi}{4}, \frac{\pi}{2}\right)$ = Range

So, $\tan^{-1}(x^2 + 2x + \alpha) \in \left[-\frac{\pi}{4}, \frac{\pi}{2}\right) \Rightarrow \quad x^2 + 2x + \alpha \in \left[\tan\left(-\frac{\pi}{4}\right), \tan\frac{\pi}{2}\right)$

$\Rightarrow \quad x^2 + 2x + \alpha \in [-1, \infty)$

$\Rightarrow$ Range of $x^2 + 2x + \alpha$ is $[-1, \infty)$

$\Rightarrow \quad -\frac{D}{4a} = -1$ (Range of quadratic (a>0) is $\left[-\frac{D}{4a}, \infty\right)$)

$\Rightarrow \quad -\frac{(4 - 4\alpha)}{4(1)} = -1$

$\Rightarrow \quad \alpha = 0$

Q22 **If $(\sin^{-1} x)^2 + (\sin^{-1} y)^2 + (\sin^{-1} z)^2 = \frac{3\pi^2}{4}$, then tell distinct possible values of $x + y - z$.**

A22 Range of $\sin^{-1}x$ is $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

Range of $(\sin^{-1}x)^2$ is $\left[0, \frac{\pi^2}{4}\right]$

Maximum value of L.H.S. $= \frac{\pi^2}{4} + \frac{\pi^2}{4} + \frac{\pi^2}{4}$

$= \frac{3\pi^2}{4}$ = R.H.S

$\Rightarrow (\sin^{-1}x)^2 = (\sin^{-1}y)^2 = (\sin^{-1}z)^2 = \frac{\pi^2}{4}$

$\Rightarrow x = 1$ or -1, $y = 1$ or -1, $z = 1$ or -1

$\Rightarrow x + y - z \in \{1 + 1 - (-1),\ 1 + 1 - 1,\ 1 - 1 - 1,\ -1 - 1 - 1\}$

$\Rightarrow x + y - z \in \{3, 1, -1, -3\}$

Q23 **Tell domain of** $\log_{10}(1-(\log_7(x^2-5x+13)))+\sin^{-1}\left(\frac{7}{6+\sin\left(\frac{9\pi x}{2}\right)}\right)$

A23 $\log_{10}x$ is defined is $x > 0$

$\Rightarrow \quad 1 - \log_7(x^2 - 5x + 13) > 0$

$\Rightarrow \quad 1 > \log_7(x^2 - 5x + 13)$

$\Rightarrow \quad 7^1 > x^2 - 5x + 13$

$\Rightarrow \quad x^2 - 5x + 6 < 0$

$\Rightarrow \quad (x - 2)(x - 3) < 0$

$\Rightarrow \quad x \in (2, 3)$

Also, $\log_7(x^2 - 5x + 13)$ is defined if $(x^2 - 5x + 13) > 0$

$\Rightarrow x \in R$

So, Domain of $\log_{10}(1 - \log_7(x^2 - 5x + 13))$ is $(2, 3) \cap R$

$\Rightarrow$ Domain of $\log_{10}(1 - \log_7(x^2 - 5x + 13))$ is $(2, 3)$

Now, $\sin^{-1}\left(\frac{7}{6+\sin\frac{9\pi x}{2}}\right)$ is defined if $-1 \le \frac{7}{6+\sin\frac{9\pi x}{2}} \le 1$

$\Rightarrow \quad -6 - \sin\frac{9\pi x}{2} \le 7, 7 \le 6 + \sin\frac{9\pi x}{2}$

$\Rightarrow \quad -13 \le \sin\frac{9\pi x}{2}, \quad 1 \le \sin\frac{9\pi x}{2}$

$\Rightarrow \quad \sin\frac{9\pi x}{2} = 1$

$\Rightarrow \quad \frac{9\pi x}{2} = 2n\pi + \frac{\pi}{2}, \quad$ where $n \in I$

$\Rightarrow \quad x = \frac{4n+1}{9}, \quad$ where $n \in I$

So, Domain of given expression is $(2,3) \cap \left\{\frac{4n+1}{9}\right\}, \quad n \in I$

$$= \left\{\frac{21}{9}, \frac{25}{9}\right\}$$

Q24 **Draw the graph of $\sin(\tan^{-1}x)$**

A24

I. Given function is odd function

II. Domain is R

III. Range: $\tan^{-1}x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ $\sin(\tan^{-1}x) \in \left(\sin\left(-\frac{\pi}{2}\right), \sin\frac{\pi}{2}\right)$

(As $\sin^{-1}x$ is increasing in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$)

$\in (-1, 1)$

IV. $\frac{dy}{dx} = \cos(\tan^{-1}x)\frac{d}{dx}(\tan^{-1}x)$

$= \frac{\cos(\tan^{-1}x)}{1+x^2} > 0$

$\Rightarrow$ Function is increasing.

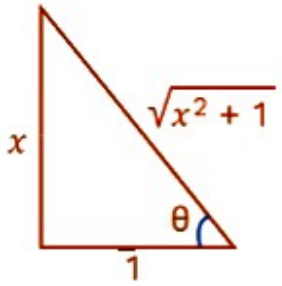
Also, Let $\tan^{-1} x = \theta$

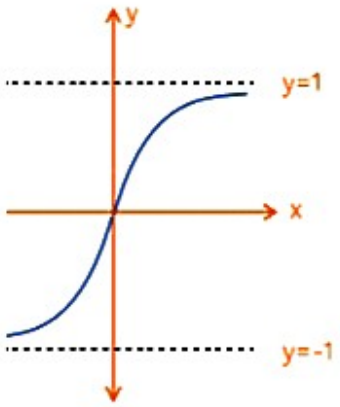
$\Rightarrow x = \tan\theta$

$f(x) = \sin(\tan^{-1} x)$

$= \sin\theta$

$= \frac{x}{\sqrt{x^2+1}}$





Properties of inverse function :

Property-I (A)

1. $\sin(\sin^{-1}x) = x, \forall\ x \in [-1, 1]$

Proof :

Let $f(x) = \sin(\sin^{-1}x)$

Domain: $x \in [-1, 1]$

Range: $y \in \sin(\sin^{-1}([-1, 1]))$

$y \in \sin\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

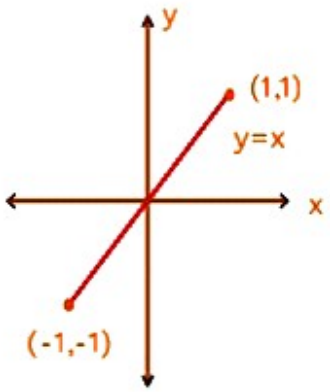
$\in [-1, 1]$

Now, Let $\sin^{-1}x = \theta \Rightarrow x = \sin\theta$

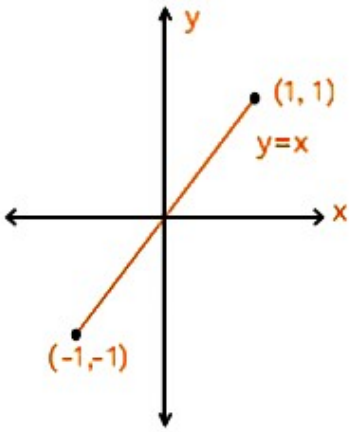
$f(x) = \sin(\sin^{-1}x)$

$= \sin(\theta)$

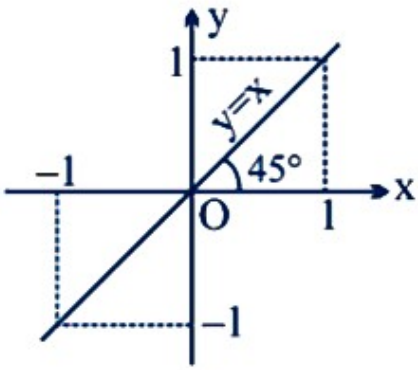
$= x$



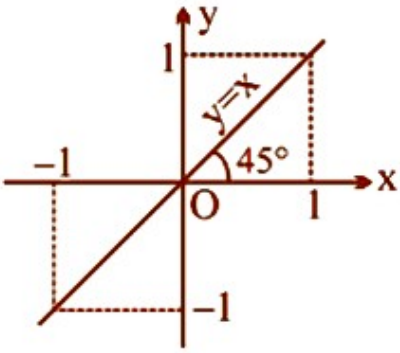
2. $\cos(\cos^{-1}(x)) = x, \forall\ x \in [-1, 1]$



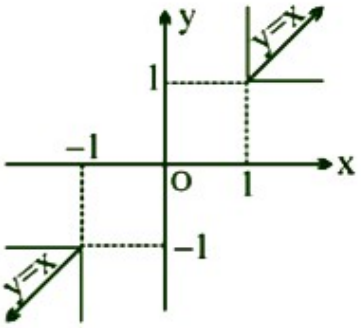
3. $\tan(\tan^{-1}x) = x, \forall\ x \in R$



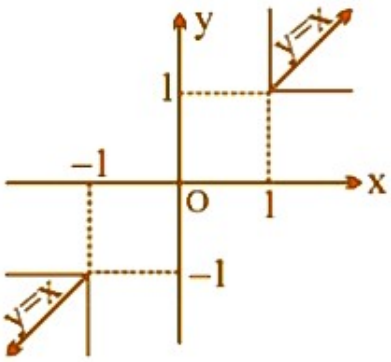
4. $\cot(\cot^{-1}x) = x, \forall\ x \in R$



5. $\text{cosec}(\text{cosec}^{-1}x) = x, \forall\ x \in (-\infty, -1] \cup [1, \infty)$

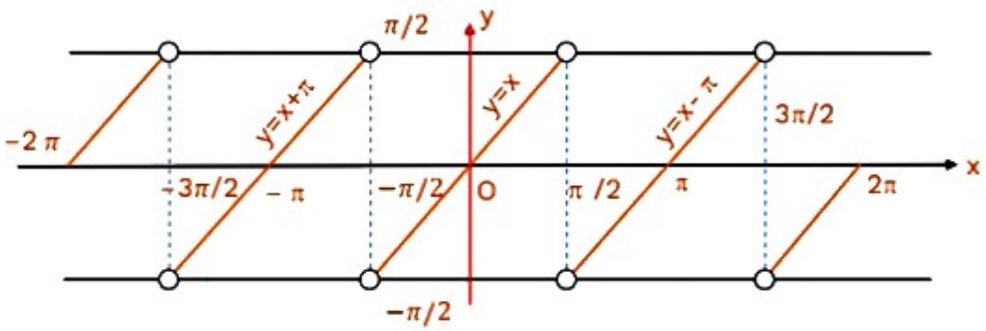


6. $\sec(\sec^{-1}x) = x, \forall\ x \in (-\infty, -1]\cup[1, \infty)$



Property I (B)

1. $f(x) = \tan^{-1}(\tan x)$



ILLUSTRATIONS :

Q1 **Tell value of** $\tan^{-1}\left(\tan\frac{13\pi}{3}\right)$

A1 $\frac{13\pi}{3} \in \left(\frac{7\pi}{2}, \frac{9\pi}{2}\right)$

$\Rightarrow \tan^{-1}\left(\tan\frac{13\pi}{3}\right) = \frac{13\pi}{3} - 4\pi = \frac{\pi}{3}$

Q2 $\tan^{-1}\left(\tan\frac{25\pi}{4}\right)$

A2 $\frac{25\pi}{4} = 6\pi + \frac{\pi}{4} \in \left(6\pi - \frac{\pi}{2}, 6\pi + \frac{\pi}{2}\right)$

$\tan^{-1}\left(\tan\frac{25\pi}{4}\right) = \frac{25\pi}{4} - 6\pi = \frac{\pi}{4}$

Q3 $\tan^{-1}\left(\tan\frac{19\pi}{4}\right)$

A3 $\frac{19\pi}{4} = 5\pi - \frac{\pi}{4} \in \left(5\pi - \frac{\pi}{2}, 5\pi + \frac{\pi}{2}\right)$

$\tan^{-1}\left(\tan\frac{19\pi}{4}\right) = \frac{19\pi}{4} - 5\pi = -\frac{\pi}{4}$

Q4 $\tan^{-1}\left(\tan\frac{27\pi}{4}\right)$

A4 $\frac{27\pi}{4} = 7\pi - \frac{\pi}{4} \in \left(7\pi - \frac{\pi}{2}, 7\pi + \frac{\pi}{2}\right)$

$\tan^{-1}\left(\tan\frac{27\pi}{4}\right) = \frac{27\pi}{4} - 7\pi = -\frac{\pi}{4}$

Q5 $\tan^{-1}\left(\tan\frac{17\pi}{3}\right)$

A5 $\frac{17\pi}{3} \in \left(6\pi - \frac{\pi}{2}, 6\pi + \frac{\pi}{2}\right)$

$\tan^{-1}\left(\tan\frac{17\pi}{3}\right) = \frac{17\pi}{3} - 6\pi = -\frac{\pi}{3}$

Q6 $\tan^{-1}(\tan 1)$

A6 $1 \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

$\Rightarrow \tan^{-1}(\tan 1) = 1$

Q7 $\tan^{-1}(\tan 2)$

A7 $2 \in \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$

$\Rightarrow \tan^{-1}(\tan 2) = 2 - \pi$

Q8 $\tan^{-1}(\tan 3)$

A8 $3 \in \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$

$\Rightarrow \tan^{-1}(\tan 3) = 3 - \pi$

Q9 $\tan^{-1}(\tan 4)$

A9 $4 \in \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$

$\Rightarrow \tan^{-1}(\tan 4) = 4 - \pi$

Q10 $\tan^{-1}(\tan 5)$

A10 $5 \in \left(\frac{3\pi}{2}, \frac{5\pi}{2}\right)$

$\Rightarrow \tan^{-1}(\tan 5) = 5 - 2\pi$

Q11 $\tan^{-1}(\tan 6)$

A11 $6 \in \left(\frac{3\pi}{2}, \frac{5\pi}{2}\right)$

$\Rightarrow \tan^{-1}(\tan 6) = 6 - 2\pi$

Q12 $\tan^{-1}(\tan 7)$

A12 $7 \in \left(\frac{3\pi}{2}, \frac{5\pi}{2}\right)$

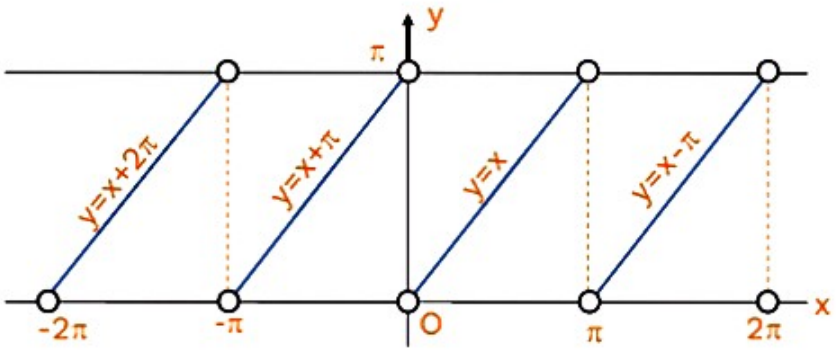
$\Rightarrow \tan^{-1}(\tan 7) = 7 - 2\pi$

Q13 $\tan^{-1}(\tan 8)$

A13 $8 \in \left(\frac{5\pi}{2}, \frac{7\pi}{2}\right)$

$\Rightarrow \quad ^{-1}(\quad 8) = 8 - 3\pi$

2. $f(x) = \cot^{-1}(\cot x)$



ILLUSTRATIONS :

Q1 $\cot^{-1}\left(\frac{11\pi}{3}\right)$

A1 $\frac{11\pi}{3} \in (3\pi, 4\pi)$

$\Rightarrow \cot^{-1}\left(\cot\frac{11\pi}{3}\right) = \frac{11\pi}{3} - 3\pi = \frac{2\pi}{3}$

Q2 $\cot^{-1}\left(\frac{17\pi}{3}\right)$

A2 $\frac{17\pi}{3} \in (5\pi, 6\pi)$

$\Rightarrow \cot^{-1}\left(\cot\frac{17\pi}{3}\right) = \frac{17\pi}{3} - 5\pi = \frac{2\pi}{3}$

Q3 $\cot^{-1}\left(\frac{13\pi}{7}\right)$

A3 $\frac{13\pi}{7} \in (\pi, 2\pi)$

$\Rightarrow \cot^{-1}\left(\cot\frac{13\pi}{7}\right) = \frac{13\pi}{7} - \pi = \frac{6\pi}{7}$

Q4 $\cot^{-1}\left(\frac{25\pi}{4}\right)$

A4 $\frac{25\pi}{4} \in (6\pi, 7\pi)$

$\Rightarrow \cot^{-1}\left(\cot\frac{25\pi}{4}\right) = \frac{25\pi}{4} - 6\pi = \frac{\pi}{4}$

Q5 **$\cot^{-1}(\cot 1)$**

A5 $1 \in (0,\ \pi)$

$\cot^{-1}(\cot 1) = 1$

Q6 $\cot^{-1}(\cot 2)$

A6 $2 \in (0,\ \pi)$

$\cot^{-1}(\cot 2) = 2$

Q7 $\cot^{-1}(\cot 3)$

A7 $3 \in (0,\ \pi)$

$\cot^{-1}(\cot 3) = 3$

Q8 $\cot^{-1}(\cot 4)$

A8 $4 \in (\pi, 2\pi)$

$\cot^{-1}(\cot 4) = 4 - \pi$

Q9 $\cot^{-1}(\cot 5)$

A9 $5 \in (\pi, 2\pi)$

$\cot^{-1}(\cot 5) = 5 - \pi$

Q10 $\cot^{-1}(\cot 6)$

A10 $6 \in (\pi, 2\pi)$

$\cot^{-1}(\cot 6) = 6 - \pi$

Q11 $\cot^{-1}(\cot 7)$

A11 $7 \in (2\pi, 3\pi)$

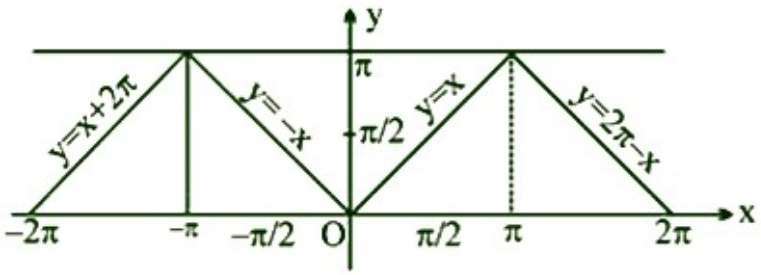
$\cot^{-1}(\cot 7) = 7 - 2\pi$

Q12 $\cot^{-1}(\cot 8)$

A12 $8 \in (2\pi, 3\pi)$

$\cot^{-1}(\cot 8) = 8 - 2\pi$

3. $f(x) = \cos^{-1}(\cos x)$



Properties :

(i) f(x) is an even function
(ii) Domain is R
(iii) Range is [0, π]
(iv) Period is 2π.

ILLUSTRATIONS:

Q1 $\cos^{-1}\left(\cos\frac{11\pi}{3}\right)$

A1 $\frac{11\pi}{3} \in (3\pi, 4\pi)$

$\cos^{-1}\left(\cos\frac{11\pi}{3}\right) = 4\pi - \frac{11\pi}{3} = \frac{\pi}{3}$

Q2 $\cos^{-1}\left(\cos\frac{25\pi}{4}\right)$

A2 $\frac{25\pi}{4} \in (6\pi, 7\pi)$

$\cos^{-1}\left(\cos\frac{25\pi}{4}\right) = \frac{25\pi}{4} - 6\pi = \frac{\pi}{4}$

Q3 $\cos^{-1}\left(\cos\frac{7\pi}{3}\right)$

A3 $\frac{7\pi}{3} \in (2\pi, 3\pi)$

$\cos^{-1}\left(\cos\frac{7\pi}{3}\right) = \frac{7\pi}{3} - 2\pi = \frac{\pi}{3}$

Q4 **$\cos^{-1}(\cos 1)$**

A4 $1 \in (0, \pi)$
$\cos^{-1}(\cos 1) = 1$

Q5 **$\cos^{-1}(\cos 2)$**

A5 $2 \in (0, \pi)$
$\cos^{-1}(\cos 2) = 2$

Q6 **$\cos^{-1}(\cos 3)$**

A6 $3 \in (0, \pi)$
$\cos^{-1}(\cos 3) = 3$

Q7 **$\cos^{-1}(\cos 4)$**

A7 $4 \in (\pi, 2\pi)$
$\cos^{-1}(\cos 4) = 2\pi - 4$

Q8 **$\cos^{-1}(\cos 5)$**

A8 $5 \in (\pi, 2\pi)$
$\cos^{-1}(\cos 5) = 2\pi - 5$

Q9 **$\cos^{-1}(\cos 6)$**

A9 $6 \in (\pi, 2\pi)$
$\cos^{-1}(\cos 6) = 2\pi - 6$

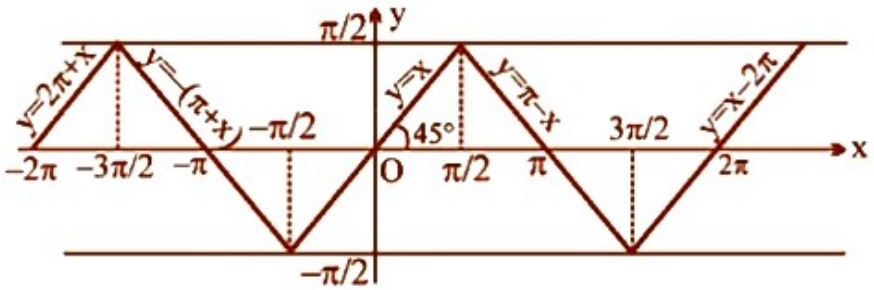
Q10 $\cos^{-1}(\cos 7)$

A10 $7 \in (2\pi, 3\pi)$
$\cos^{-1}(\cos 7) = 7 - 2\pi$

Q10 $\cos^{-1}(\cos 8)$

A10 Sol. $8 \in (2\pi, 3\pi)$
$\cos^{-1}(\cos 8) = 8 - 2\pi$

4. $f(x) = \sin^{-1}(\sin x)$



Properties :

(i) f(x) is odd function
(ii) Domain is R
(iii) Range is $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
(iv) Period is 2π

ILLUSTRATIONS :

Q1 $\sin^{-1}\left(\sin\frac{13\pi}{3}\right)$

A1 $\frac{13\pi}{3} \in \left(\frac{7\pi}{2}, \frac{9\pi}{2}\right)$

$\sin^{-1}\left(\sin\frac{13\pi}{3}\right) = \frac{13\pi}{3} - 4\pi = \frac{\pi}{3}$

Q2 $\sin^{-1}\left(\sin\frac{25\pi}{4}\right)$

A2 $\frac{25\pi}{4} \in \left(\frac{11\pi}{2}, \frac{13\pi}{2}\right)$

$\sin^{-1}\left(\sin\frac{25\pi}{4}\right) = \frac{25\pi}{4} - 6\pi = \frac{\pi}{4}$

Q3 $\sin^{-1}\left(\sin\frac{19\pi}{4}\right)$

A3 $\sin^{-1}\left(\sin\left(4\pi + \frac{3\pi}{4}\right)\right) = \sin^{-1}\left(\sin\frac{3\pi}{4}\right) = \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) = \frac{\pi}{4}$

Q4 $\sin^{-1}\left(\sin\frac{37\pi}{4}\right)$

A4 $\frac{37\pi}{4} \in \left(\frac{17\pi}{2}, \frac{19\pi}{2}\right)$

$\sin^{-1}\left(\sin\frac{37\pi}{4}\right) = 9\pi - \frac{37\pi}{4} = -\frac{\pi}{4}$

Q5 $\sin^{-1}\left(\sin\frac{17\pi}{3}\right)$

A5 $\frac{17\pi}{3} \in \left(\frac{11\pi}{2}, \frac{13\pi}{2}\right)$

$\sin^{-1}\left(\sin\frac{17\pi}{3}\right) = \frac{17\pi}{3} - 6\pi$

$= -\frac{\pi}{3}$

Q6 $\sin^{-1}(\sin 1)$

A6 $1 \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

$\sin^{-1}(\sin 1) = 1$

Q7 $\sin^{-1}(\sin 2)$

A7 $2 \in \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$

$\sin^{-1}(\sin 2) = 2 - \pi$

Q8 $\sin^{-1}(\sin 3)$

A8 $3 \in \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$

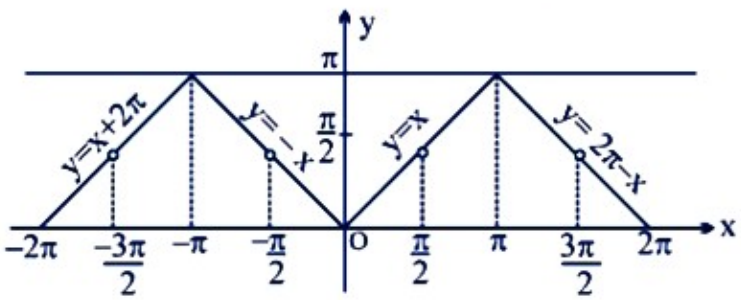
$\sin^{-1}(\sin 3) = 3 - \pi$

Q9 $\sin^{-1}(\sin 8)$

A9 $8 \in \left(\frac{5\pi}{2}, \frac{7\pi}{2}\right)$

$\sin^{-1}(\sin 8) = 3\pi - 8$

5. $f(x) = \sec^{-1}(\sec x)$



Properties :

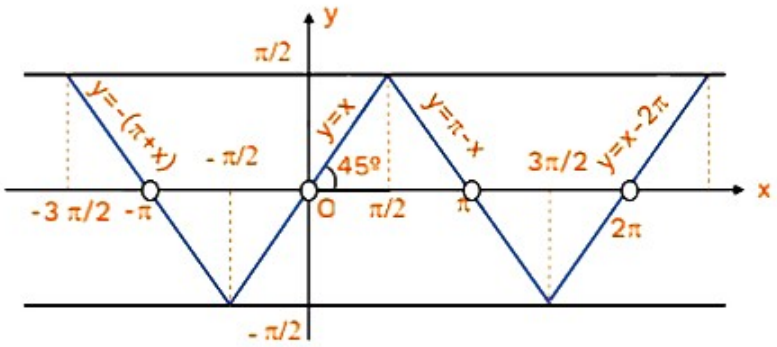
(i) f(x) is an even function

(ii) Domain is $R-(2n+1)\frac{\pi}{2}$, $n \in I$

(iii) Range is $[0, \pi]-\left\{\frac{\pi}{2}\right\}$

(iv) Period is 2π.

6. $\text{cosec}^{-1}(\text{cosec}x)$



Properties :

(i) f(x) is odd function

(ii) Domain is $R-\{n\pi\}$, $n \in I$

(iii) Range is $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]-\{0\}$

(iv) Period is 2π.

Inequality Based Illustrations :

Q1 **Solve for x : $\log_2(\tan^{-1}x) > 1$**

A1 $\log_2(\tan^{-1}x) > \log_2 2$

$\Rightarrow (\tan^{-1}x) > 2$ (As $\log_2 x$ is increasing function)

$\Rightarrow x \in \phi$ (As range of $\tan^{-1}x$ is $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$)

Q2 **Solve for x: $(\cot^{-1}x)^2 - 5(\cot^{-1}x) + 6 > 0$**

A2 $(\cot^{-1}x) = u$
$\Rightarrow u^2 - 5u + 6 > 0$
$\Rightarrow (u - 2)(u - 3) > 0$
$\Rightarrow u \in (-\infty, 2) \cup (3, \infty)$
$\Rightarrow \cot^{-1}x \in (0, 2) \cup (3, \pi)$ (As $\cot^{-1}x \in (0, \pi)$)
$\Rightarrow x \in (\cot 2, \cot 0) \cup (\cot\pi, \cot 3)$ (As $\cot^{-1}x$ is decreasing function)
$\Rightarrow x \in (\cot 2, \infty) \cup (-\infty, \cot 3)$

Q3 **Solve for x: $0 \le \cos^{-1}x < 1$**

A3 $\Rightarrow \cos 1 < x \le \cos 0$ (As $\cos^{-1}x$ is decreasing function)
$\Rightarrow \cos 1 < x \le 1$

Q4 **Tell domain of** $f(x) = \dfrac{1}{\sqrt{\ln(\cot^{-1}x)}}$

A4 $\ln(\cot^{-1}x) > 0$
$\Rightarrow \cot^{-1}x > 1$
$\Rightarrow x < \cot 1$ (As $\cot^{-1}x$ is decreasing function)
$\Rightarrow x \in (-\infty, \cot 1)$

Property-II:

1. (i) $\operatorname{cosec}^{-1}x = \sin^{-1}\dfrac{1}{x}, \forall x \in (-\infty, -1] \cup [1, \infty)$

(ii) $\sin^{-1}x = \operatorname{cosec}^{-1}\dfrac{1}{x}, \forall x \in [-1, 1] - \{0\}$

Proof :

Let $\text{cosec}^{-1}x = \theta$, where $\theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$

$\Rightarrow x = \text{cosec}\theta$

$\Rightarrow \frac{1}{x} = \sin\theta$

$\Rightarrow \theta = \sin^{-1}\left(\frac{1}{x}\right)$

2. (i) $\sec^{-1}x = \cos^{-1}\frac{1}{x}, \forall x \in (-\infty, -1] \cup [1, \infty)$

(ii) $\cos^{-1}x = \sec^{-1}\frac{1}{x}, \forall x \in [-1, 1] - \{0\}$

3. $\cot^{-1}x = \begin{cases} \tan^{-1}\frac{1}{x} & x > 0 \\ \pi + \tan^{-1}\frac{1}{x} & x < 0 \end{cases}$

Proof :

$\cot^{-1}x = \theta$, where $\theta \in (0, \pi)$

$\Rightarrow x = \cot\theta$

$\Rightarrow \frac{1}{x} = \tan\theta$

$\Rightarrow \tan^{-1}\left(\frac{1}{x}\right) = \tan^{-1}(\tan\theta)$

$$= \begin{cases} \theta = \cot^{-1}x & 0 < \theta < \frac{\pi}{2} \Rightarrow x > 0 \\ \theta - \pi = \cot^{-1}x - \pi & \frac{\pi}{2} < \theta < \pi \Rightarrow x < 0 \end{cases}$$

$\Rightarrow \tan^{-1}\frac{1}{x} = \cot^{-1}x$ when $x > 0$ and $\tan^{-1}\frac{1}{x} = \cot^{-1}x - \pi$ when $x < 0$

$\Rightarrow \cot^{-1}x = \tan^{-1}\frac{1}{x}$ when $x > 0$ and $\cot^{-1}x = \pi + \tan^{-1}\frac{1}{x}$ when $x < 0$

Property-III :

1. $\sin^{-1}(-x) = -\sin^{-1}x$
2. $\text{cosec}^{-1}(-x) = -\text{cosec}^{-1}(x)$
3. $\cos^{-1}(-x) = \pi - \cos^{-1}(x)$
4. $\sec^{-1}(-x) = \pi - \sec^{-1}(x)$
5. $\tan^{-1}(-x) = -\tan^{-1}(x)$
6. $\cot^{-1}(-x) = \pi - \cot^{-1}(x)$

Proof of 3 :

$\cos^{-1}x = \theta, \quad \theta \in [0, \pi]$

$x = \cos\theta$

$-x = \cos(\pi - \theta)$

$\cos^{-1}(-x) = \cos^{-1}(\cos(\pi - \theta))$

$= \pi - \theta \qquad (\text{As } \pi - \theta \in [0, \pi])$

$= \pi - \cos^{-1}x$

Property-IV :

1. $\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}, \ \forall\ x \in [-1, 1]$
2. $\tan^{-1}x + \cot^{-1}x = \frac{\pi}{2}, \ \forall\ x \in R$
3. $\sec^{-1}x + \text{cosec}^{-1}x = \frac{\pi}{2},$

   $\forall\ x \in (-\infty, -1] \cup [1, \infty)$

Proof of 1 :

Let $\sin^{-1}x = \theta, \ \theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

$\Rightarrow x = \sin\theta$

$\Rightarrow x = \cos\left(\frac{\pi}{2} - \theta\right)$

$\Rightarrow \cos^{-1}x = \cos^{-1}\left(\cos\left(\frac{\pi}{2} - \theta\right)\right)$

Now, $\theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \Rightarrow -\theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \Rightarrow \frac{\pi}{2} - \theta \in [0, \pi]$

So, $\cos^{-1}x = \frac{\pi}{2} - \theta$

$\Rightarrow \sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}$

ILLUSTRATIONS :

Q1 **Tell value of** $\tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}(2) + \tan^{-1}\left(\sqrt{3}\right)$

A1 $\cot^{-1}(2) + \tan^{-1}(2) + \frac{\pi}{3}$

$\Rightarrow \frac{\pi}{2} + \frac{\pi}{3} = \frac{5\pi}{6}$

Q2 **Tell x if** $\sin^{-1}x + \cos^{-1}\left(x^2 - 2x + 2\right) = \frac{\pi}{2}$

A2 Domain: $(x^2 - 2x + 2) \leq 1$

$\Rightarrow (x - 1)^2 + 1 \leq 1$

$\Rightarrow (x - 1)^2 \leq 0$

$\Rightarrow x = 1$ (which is also satisfying $-1 \leq x \leq 1$)

Also, $x^2 - 2x + 2 = x$

$\Rightarrow x^2 - 3x + 2 = 0$

$\Rightarrow x = 1, 2$

$\Rightarrow x = 1$ (As $x = 2$ is not in domain)

Q3 If $\sin^{-1}a + \cos^{-1}b = \frac{\pi}{2}$ and $\sec^{-1}a + \operatorname{cosec}^{-1}b = \frac{\pi}{2}$ then find a + b

(A) 2 **(B) –2** **(C) 0** **(D) None of these**

A3 **(AB)**

$\sin^{-1}a + \cos^{-1}b = \frac{\pi}{2} \Rightarrow a = b, -1 \le a, b \le 1$ .....(i)

$\operatorname{cosec}^{-1}a + \sec^{-1}b = \frac{\pi}{2}$

$\Rightarrow a = b,\ a, b \in (-\infty, -1] \cup [1, \infty)$ .....(ii)

By (i) and (ii)

$a = b = 1$ OR $a = b = -1$

$\Rightarrow a + b = 2$ OR -2

Q4 If $\left(\tan^{-1}x\right)^2 + \left(\cot^{-1}x\right)^2 = \frac{5\pi^2}{8}$, then find the x

A4 Let $\tan^{-1}x = \theta \Rightarrow \cot^{-1}x = \frac{\pi}{2} - \theta$

So, given equation becomes $\theta^2 + \left(\frac{\pi}{2} - \theta\right)^2 = \frac{5\pi^2}{8}$

$\Rightarrow 2\theta^2 - \pi\theta + \frac{\pi^2}{4} - \frac{5\pi^2}{8} = 0$

$\Rightarrow 2\theta^2 - \pi\theta - \frac{3\pi^2}{8} = 0$

$\Rightarrow 16\theta^2 - 8\pi\theta - 3\pi^2 = 0$

$\Rightarrow (4\theta + \pi)(4\theta - 3\pi) = 0$

$\Rightarrow \theta = -\frac{\pi}{4}, \frac{3\pi}{4}$

But $\theta = \frac{3\pi}{4}$ not possible as $\theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

$\Rightarrow \theta = -\frac{\pi}{4}$

$\Rightarrow \tan^{-1}x = -\frac{\pi}{4}$

$\Rightarrow x = \tan\left(-\frac{\pi}{4}\right)$ $\Rightarrow x = -1$

Q5 **Solve for x :** $5\tan^{-1}x + 3\cot^{-1}x = \frac{7\pi}{4}$

A5 $2\tan^{-1}x + 3\left(\tan^{-1}x + \cot^{-1}x\right) = \frac{7\pi}{4}$

$$\Rightarrow 2\tan^{-1}x = \frac{7\pi}{4} - \frac{3\pi}{2}$$

$$\Rightarrow \tan^{-1}x = \frac{\pi}{8}$$

$$\Rightarrow x = \tan\frac{\pi}{8}$$

$$= \sqrt{2} - 1$$

Note :

To find $\tan\frac{\pi}{8}$

$$\tan\frac{\theta}{2} = \frac{1-\cos\theta}{\sin\theta}$$

$$\tan\frac{\pi}{8} = \frac{1-\cos\frac{\pi}{4}}{\sin\frac{\pi}{4}}$$

Point to Remember!!!

$$\tan\frac{\pi}{8} = \sqrt{2} - 1$$

$$\tan\frac{3\pi}{8} = \sqrt{2} + 1$$

$$\tan\frac{\pi}{12} = 2 - \sqrt{3}$$

$$\tan\frac{5\pi}{12} = 2 + \sqrt{3}$$

ILLUSTRATION :

Q1 **Solve for x,** $4\sin^{-1}x + \cos^{-1}x = \frac{3\pi}{4}$

A1 $3\sin^{-1}x + \sin^{-1}x + \cos^{-1}x = \frac{3\pi}{4}$

$$\Rightarrow 3\sin^{-1}x + \frac{\pi}{2} = \frac{3\pi}{4}$$

$$\Rightarrow \sin^{-1}x = \frac{\pi}{12} \Rightarrow x = \sin\frac{\pi}{12}$$

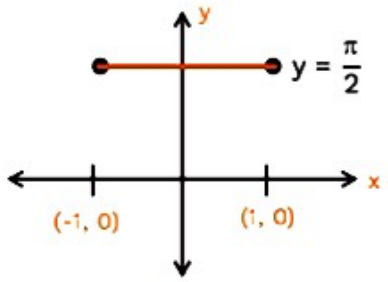
$$\Rightarrow x = \frac{\sqrt{3}-1}{2\sqrt{2}}$$

Note :

1.

Graph of $f(x) = \sin^{-1}x + \cos^{-1}x$

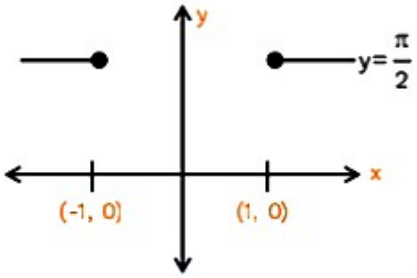
Domain: $[-1, 1]$, $f(x) = \frac{\pi}{2}$



2.

Graph of $f(x) = \sec^{-1}x + \text{cosec}^{-1}x$

Domain: $(-\infty, -1] \cup [1, \infty)$, $f(x) = \frac{\pi}{2}$



ILLUSTRATIONS :

Q1 **If $\cot^{-1}\left(\frac{n}{\pi}\right) > \frac{\pi}{6}$, then find maximum value of $n(\in N)$.**

A1 $\cot^{-1}\frac{n}{\pi} \in \left(\frac{\pi}{6}, \pi\right)$

$\Rightarrow \frac{n}{\pi} \in \left(\cot\pi, \cot\frac{\pi}{6}\right)$ (As graph of $\cot^{-1}x$ is decreasing)

$\Rightarrow \frac{n}{\pi} \in \left(-\infty, \sqrt{3}\right)$

$\Rightarrow \frac{n}{\pi} < \sqrt{3}$

$\Rightarrow n < \pi\sqrt{3}$ (approximately 5.44)

Greatest possible integral n = 5.

Q2 **Tell maximum and minimum value of $\left(\sin^{-1}x\right)^3 + \left(\cos^{-1}x\right)^3$**

A2 Let $\sin^{-1}x = \theta \Rightarrow \cos^{-1}x = \frac{\pi}{2} - \theta$

Now, $y = \theta^3 + \left(\frac{\pi}{2} - \theta\right)^3$

Use $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$

$$y = \left(\theta + \frac{\pi}{2} - \theta\right)\left[\theta^2 + \left(\frac{\pi}{2} - \theta\right)^2 - \theta\left(\frac{\pi}{2} - \theta\right)\right]$$

$$= \frac{\pi}{2}\left(3\theta^2 - \frac{3\pi}{2}\theta + \frac{\pi^2}{4}\right)$$

$$= \frac{3\pi}{2}\left(\theta^2 - \frac{\pi}{2}\theta + \frac{\pi^2}{12}\right)$$

$$= \frac{3\pi}{2}\left[\left(\theta - \frac{\pi}{4}\right)^2 + \frac{\pi^2}{12} - \frac{\pi^2}{16}\right]$$

$$= \frac{3\pi}{2}\left[\left(\theta - \frac{\pi}{4}\right)^2 + \frac{\pi^2}{48}\right]$$

Now, $\theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ $\quad \theta - \frac{\pi}{4} \in \left[-\frac{3\pi}{4}, \frac{\pi}{4}\right]$

$$\left(\theta - \frac{\pi}{4}\right)^2 \in \left[0, \frac{9\pi^2}{16}\right]$$

$$\left(\theta - \frac{\pi}{4}\right)^2 + \frac{\pi^2}{48} \in \left[\frac{\pi^2}{48}, \frac{7\pi^2}{12}\right]$$

$$\frac{3\pi}{2}\left[\left(\theta - \frac{\pi}{4}\right)^2 + \frac{\pi^2}{48}\right] \in \left[\frac{\pi^3}{32}, \frac{7\pi^3}{8}\right]$$

Maximum value = $\frac{7\pi^3}{8}$; Minimum value = $\frac{\pi^3}{32}$

Q3 **Find range of** $f(x) = \sin^{-1}x + \cos^{-1}x + \tan^{-1}x$

A3 Domain: $[-1, 1]$

$f(x) = \frac{\pi}{2} + \tan^{-1}x$ (as $\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}$)

$f(x) \in \frac{\pi}{2} + \left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$ (as $\tan^{-1}x$ is increasing in $[-1, 1]$)

$f(x) \in \left[\frac{\pi}{4}, \frac{3\pi}{4}\right]$

Q4 **Tell range of** $f(x) = \sec^{-1}x + \operatorname{cosec}^{-1}x + \cot^{-1}x$

A4 Domain $(-\infty, -1] \cup [1, \infty)$

$f(x) = \frac{\pi}{2} + \cot^{-1}x$ (as $\sec^{-1}x + \operatorname{cosec}^{-1}x = \frac{\pi}{2}$)

$f(x) \in \frac{\pi}{2} + \left[\frac{3\pi}{4}, \pi\right) \cup \left(0, \frac{\pi}{4}\right]$ (as $\cot^{-1}x$ is decreasing in $(-\infty, -1] \cup [1, \infty)$)

$f(x) \in \left(\frac{\pi}{2}, \frac{3\pi}{4}\right] \cup \left[\frac{5\pi}{4}, \frac{3\pi}{2}\right)$

Q5 **Tell domain and range of** $f(x) = \sin^{-1}x + \tan^{-1}x + \sec^{-1}x$

A5 Domain: $x \in [-1,1] \cap ((-\infty,-1] \cup [1,\infty)) \Rightarrow x \in \{-1, 1\}$

Range = $\{f(1), f(-1)\} = \left\{\frac{\pi}{2}+\frac{\pi}{4}+0, -\frac{\pi}{2}-\frac{\pi}{4}+\pi\right\} = \left\{\frac{\pi}{4}, \frac{3\pi}{4}\right\}$

Q6 **Number of solution(s) of equation** $\tan^{-1}\sqrt{x^2+x} + \sin^{-1}\sqrt{x^2+x+1} = \frac{\pi}{2}$

A6 Domain: $x^2 + x \geq 0$ ...(i) (Because it is inside square root)

Also $-1 \leq \sqrt{x^2+x+1} \leq 1$ (Because it is inside $\sin^{-1}$)

$\Rightarrow \sqrt{x^2+x+1} \leq 1$ (As square root is always greater than negative)

Squaring

$\Rightarrow x^2 + x \leq 0$ ...(ii)

By (i) and (ii):

$x^2 + x = 0$

$\Rightarrow \quad x(x+1) = 0$

$\Rightarrow \quad x = 0, -1$

Domain = {0, –1}

Now check by putting 0 : $\tan^{-1}0 + \sin^{-1}1 = \frac{\pi}{2}$ (satisfied)

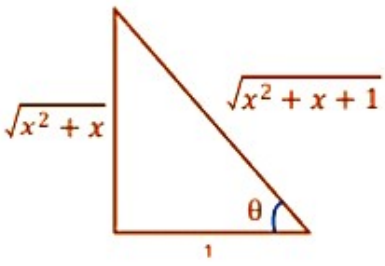
check by putting – 1 : $\tan^{-1}0 + \sin^{-1}1 = \frac{\pi}{2}$ (satisfied)

Hence, {–1, 0}

So, number of solutions = 2.

Method-II

Let $\tan^{-1}\sqrt{x^2+x} = \theta \Rightarrow \sqrt{x^2+x} = \tan\theta$



$\cos\theta = \frac{1}{\sqrt{x^2+x+1}}$

Given equation: $\cos^{-1}\frac{1}{\sqrt{x^2+x+1}} + \sin^{-1}\sqrt{x^2+x+1} = \frac{\pi}{2}$

$\Rightarrow \frac{1}{\sqrt{x^2+x+1}} = \sqrt{x^2+x+1} \Rightarrow x^2+x+1 = 1 \Rightarrow x = -1, 0$

But, still we need to check whether solutions are in domain or not.
Here, both are in domain.
So, Number of solutions = 2

IMPORTANT RESULT:

Point to Remember!!!

$$\tan^{-1}x + \tan^{-1}\frac{1}{x} = \begin{cases} \frac{\pi}{2} & x > 0 \\ -\frac{\pi}{2} & x < 0 \end{cases}$$

Proof :

For $x > 0$;

$$\tan^{-1}\frac{1}{x} = \cot^{-1}x$$

$$\Rightarrow \tan^{-1}x + \tan^{-1}\frac{1}{x} = \tan^{-1}x + \cot^{-1}x = \frac{\pi}{2}$$

For $x < 0$; $\tan^{-1}x + \tan^{-1}\frac{1}{x}$

$$= -\left[\tan^{-1}(-x) + \tan^{-1}\left(-\frac{1}{x}\right)\right]$$

$$= -\frac{\pi}{2}$$

ILLUSTRATIONS :

Q1 **If α and β are roots of equation $x^2 + 7x - 8 = 0$. Find the value of**

$$\tan^{-1}\alpha + \tan^{-1}\frac{1}{\alpha} + \tan^{-1}\beta + \tan^{-1}\frac{1}{\beta}$$

A1 Here, $\alpha\beta < 0$

$\Rightarrow \alpha$ and β are of opposite sign.

Let $\alpha > 0$, $\beta < 0$

$$\tan^{-1}\alpha + \tan^{-1}\frac{1}{\alpha} + \tan^{-1}\beta + \tan^{-1}\frac{1}{\beta} = \frac{\pi}{2} + \left(-\frac{\pi}{2}\right) = 0$$

Q2 **Tell true or false** $f(x) = \tan\left(\cot^{-1}x\right)$ and $g(x) = \cot\left(\tan^{-1}x\right)$ **are identical functions.**

A2 Domain of $f(x): \cot^{-1}x \neq \frac{\pi}{2} \Rightarrow x \neq \cot\frac{\pi}{2}$
$\Rightarrow x \neq 0$

Domain of $g(x): \tan^{-1}x \neq 0$

$\Rightarrow x \neq \tan 0 \Rightarrow x \neq 0$

Hence, Domain of f(x) = Domain of g(x) ... (i)

Now, $f(x) = \tan\left(\cot^{-1}x\right) = \tan\left(\frac{\pi}{2} - \tan^{-1}x\right) = \cot\left(\tan^{-1}(x)\right) = g(x)$

$\Rightarrow f(x) = g(x)$ ... (ii)

By (i) and (ii): f(x) and g(x) are identical functions.

True

Q3 **Solve for x :** $\cot^{-1}\left(\frac{x^2-1}{2x}\right) + \tan^{-1}\left(\frac{2x}{x^2-1}\right) = \frac{2\pi}{3}$

A3 **Case-I :** Let $\frac{x^2-1}{2x} > 0$

For $x > 0$: $\cot^{-1}x = \tan^{-1}\frac{1}{x}$

Given equation becomes : $\tan^{-1}\left(\frac{2x}{x^2-1}\right) + \tan^{-1}\left(\frac{2x}{x^2-1}\right) = \frac{2\pi}{3}$

$\Rightarrow \tan^{-1}\left(\frac{2x}{x^2-1}\right) = \frac{\pi}{3}$

$\Rightarrow \frac{2x}{x^2-1} = \tan\frac{\pi}{3}$

$\Rightarrow \frac{2x}{x^2-1} = \sqrt{3}$

$\Rightarrow \sqrt{3}x^2 - 2x - \sqrt{3} = 0$

$\Rightarrow \left(x - \sqrt{3}\right)\left(\sqrt{3}x + 1\right) = 0$

$\Rightarrow x = \sqrt{3}, -\frac{1}{\sqrt{3}}$

Both are satisfying $\frac{x^2-1}{2x} > 0$

$\Rightarrow$ Both $x = \left\{\sqrt{3}, -\frac{1}{\sqrt{3}}\right\}$ are solutions.

Case-II : Let $\frac{x^2-1}{2x} < 0$

For $x < 0$: $\cot^{-1}x = \pi + \tan^{-1}\frac{1}{x}$

Given equation becomes: $\tan^{-1}\left(\frac{2x}{x^2-1}\right) + \pi + \tan^{-1}\left(\frac{2x}{x^2-1}\right) = \frac{2\pi}{3}$

$\Rightarrow 2\tan^{-1}\left(\frac{2x}{x^2-1}\right) = \frac{2\pi}{3} - \pi$

$\Rightarrow \tan^{-1}\left(\frac{2x}{x^2-1}\right) = -\frac{\pi}{6}$

$\Rightarrow \frac{2x}{x^2-1} = -\frac{1}{\sqrt{3}}$

$\Rightarrow x^2 - 1 = -2\sqrt{3}x$

$\Rightarrow x^2 + 2\sqrt{3}x - 1 = 0$

$\Rightarrow x = -\sqrt{3} - 2, -\sqrt{3} + 2$ Both are satisfying $\frac{x^2-1}{2x} < 0$

$\Rightarrow x = \left\{-\sqrt{3} - 2, -\sqrt{3} + 2\right\}$ are solutions.

Hence, solutions are $\left\{\sqrt{3}, -\frac{1}{\sqrt{3}}, -\sqrt{3} - 2, -\sqrt{3} + 2\right\}$

Property-V :

$$\tan^{-1}x + \tan^{-1}y = \begin{cases} \tan^{-1}\left(\frac{x+y}{1-xy}\right), & x, y > 0, xy < 1 \\ \pi + \tan^{-1}\left(\frac{x+y}{1-xy}\right), & x, y > 0, xy > 1 \\ \frac{\pi}{2}, & x, y > 0, xy = 1 \end{cases}$$

$$\tan^{-1}x - \tan^{-1}y = \tan^{-1}\frac{x-y}{1+xy},$$

$x, y > 0$

Know the facts



Memorizing trick:

$$\tan(\theta + \alpha) = \frac{\tan\theta + \tan\alpha}{1 - \tan\theta\tan\alpha}$$

$$\tan(\pi + \theta) = \tan\theta$$

Proof :

Let $\tan^{-1}x = A,\ \tan^{-1}y = B,\ A,\ B \in \left(0, \frac{\pi}{2}\right)$

$$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

Take $\tan^{-1}$ on both sides

$$\tan^{-1}\tan(A+B) = \tan^{-1}\left(\frac{\tan A + \tan B}{1 - \tan A \tan B}\right)$$

$$\tan^{-1}\tan(A+B) = \tan^{-1}\left(\frac{x+y}{1-xy}\right) \qquad ...(i)$$

Here, $A + B \in \left(0, \frac{\pi}{2}\right) + \left(0, \frac{\pi}{2}\right) \in (0, \pi)$

Case-I :

$$A + B \in \left(0, \frac{\pi}{2}\right) \Rightarrow \tan^{-1}x + \tan^{-1}y < \frac{\pi}{2} \Rightarrow \tan^{-1}x < \cot^{-1}y \Rightarrow x < \frac{1}{y} \Rightarrow xy < 1$$

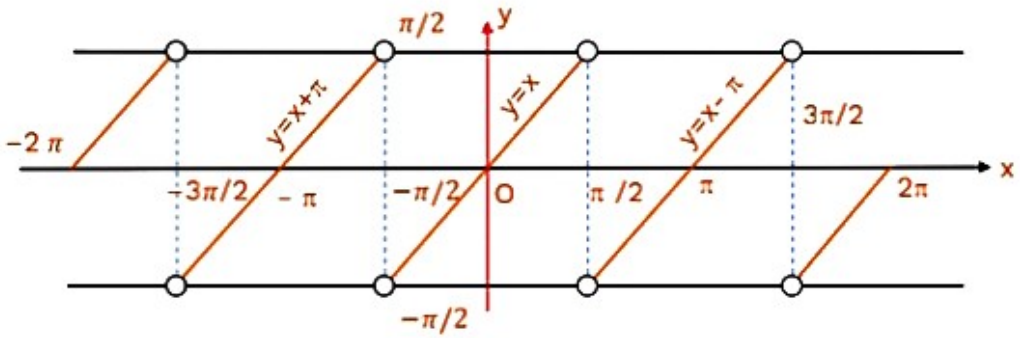
Equation (i): $A + B = \tan^{-1}\left(\frac{x+y}{1-xy}\right)$

$$\tan^{-1}x + \tan^{-1}y = \tan^{-1}\left(\frac{x+y}{1-xy}\right)$$

Case-II :

$$A + B \in \left(\frac{\pi}{2}, \pi\right) \Rightarrow \tan^{-1}x + \tan^{-1}y > \frac{\pi}{2} \Rightarrow \tan^{-1}x > \cot^{-1}y \Rightarrow x > \frac{1}{y} \Rightarrow xy > 1$$

Using graph of $y = \tan^{-1}(\tan x)$



Equation (i) becomes $A + B - \pi = \tan^{-1}\left(\frac{x+y}{1-xy}\right)$

$$\tan^{-1}x + \tan^{-1}y = \pi + \tan^{-1}\left(\frac{x+y}{1-xy}\right)$$

Q1 **Tell value of $\tan^{-1}1 + \tan^{-1}2 + \tan^{-1}3$**

A1 $2 \times 3 = 6 > 1$

So, $\tan^{-1}2 + \tan^{-1}3 = \pi + \tan^{-1}\left(\frac{2+3}{1-2\times3}\right)$

$$\Rightarrow \tan^{-1}1 + \tan^{-1}2 + \tan^{-1}3 = \frac{\pi}{4} + \pi + \tan^{-1}(-1)$$

$$= \frac{\pi}{4} + \pi - \frac{\pi}{4} = \pi$$

Q2 **Tell value of** $\tan^{-1}1 + \tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{3}$

A2 $\frac{1}{2}\times\frac{1}{3} = \frac{1}{6} < 1$

So, $\tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{3} = \tan^{-1}\left(\frac{\frac{1}{2}+\frac{1}{3}}{1-\frac{1}{2}\times\frac{1}{3}}\right)$

$$\Rightarrow \tan^{-1}1 + \tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{3} = \frac{\pi}{4} + \tan^{-1}(1)$$

$$= \frac{\pi}{4} + \frac{\pi}{4}$$

$$= \frac{\pi}{2}$$

Shortcut :

If $\tan x \tan y + \tan y \tan z + \tan z \tan x = 1$, then $x + y + z = (2n + 1)\frac{\pi}{2}$, $n \in I$.

Here, Let $x = \tan^{-1}1$, $y = \tan^{-1}\left(\frac{1}{2}\right)$, $z = \tan^{-1}\left(\frac{1}{3}\right)$

$\Rightarrow \tan x = 1$, $\tan y = \frac{1}{2}$, $\tan z = \frac{1}{3}$

Here, $\tan x \tan y + \tan y \tan z + \tan z \tan x = 1$

$\Rightarrow x + y + z = (2n+1)\frac{\pi}{2}$, $n \in I$

But $x, y, z \in \left(0, \frac{\pi}{2}\right)$

$\Rightarrow x + y + z = \frac{\pi}{2}$

Q3 **Show that** $\frac{\tan^{-1}1 + \tan^{-1}2 + \tan^{-1}3}{\cot^{-1}1 + \cot^{-1}2 + \cot^{-1}3} = 2$

A3 $\text{LHS} = \frac{\tan^{-1}1 + \tan^{-1}2 + \tan^{-1}3}{\tan^{-1}1 + \tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{3}}$ (Using $\cot^{-1}x = \tan^{-1}\left(\frac{1}{x}\right), x > 0$)

$= \frac{\pi}{\pi/2} = 2 = \text{RHS}$ (Using previous questions)

Q4 **If** $\tan^{-1}4 + \tan^{-1}5 = \cot^{-1}(\lambda)$ **then find** λ

A4 As $4 \times 5 > 1$

$\text{LHS} = \pi + \tan^{-1}\left(\frac{4+5}{1-4\times5}\right) = \pi + \tan^{-1}\left(-\frac{9}{19}\right)$

$= \pi + \left(\cot^{-1}\left(-\frac{19}{9}\right) - \pi\right)$

$= \cot^{-1}\left(-\frac{19}{9}\right)$

$\lambda = -\frac{19}{9}$

Q5 If $\alpha = \tan^{-1}5 - \tan^{-1}3 + \tan^{-1}\frac{7}{9}$ and $\beta = \tan^{-1}\frac{2}{11} + \cot^{-1}\frac{24}{7} + \tan^{-1}\frac{1}{3}$, **then**

(A) $\alpha = \beta$ **(B)** $\alpha > \beta$

(C) $\alpha < \beta$ **(D)** $\alpha + \beta = \frac{\pi}{2}$

A5 **(AD)**

$$\alpha = \tan^{-1}5 - \tan^{-1}3 + \tan^{-1}\frac{7}{9}$$

$$= \tan^{-1}\left(\frac{5-3}{1+5\times3}\right) + \tan^{-1}\frac{7}{9}$$

$$= \tan^{-1}\left(\frac{1}{8}\right) + \tan^{-1}\left(\frac{7}{9}\right)$$

$$= \tan^{-1}\left(\frac{\frac{1}{8}+\frac{7}{9}}{1-\frac{1}{8}\times\frac{7}{9}}\right)$$

$$= \tan^{-1}(1)$$

$$= \frac{\pi}{4}$$

$$\beta = \tan^{-1}\frac{2}{11} + \cot^{-1}\frac{24}{7} + \tan^{-1}\frac{1}{3}$$

$$= \tan^{-1}\frac{2}{11} + \tan^{-1}\frac{7}{24} + \tan^{-1}\frac{1}{3} \qquad \left(\text{Using } \cot^{-1}x = \tan^{-1}\frac{1}{x},\ x > 0\right)$$

$$= \tan^{-1}\left(\frac{\frac{2}{11}+\frac{7}{24}}{1-\frac{2}{11}\times\frac{7}{24}}\right) + \tan^{-1}\frac{1}{3}$$

$$= \tan^{-1}\left(\frac{125}{250}\right) + \tan^{-1}\frac{1}{3}$$

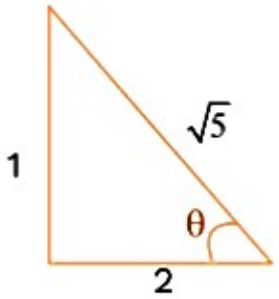
$$= \tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\frac{1}{3}$$

$$= \tan^{-1}\left(\frac{\frac{1}{2}+\frac{1}{3}}{1-\frac{1}{2}\times\frac{1}{3}}\right)$$

$$= \tan^{-1}(1) = \frac{\pi}{4}$$

Q6 **Find 'x' satisfying** $\sin^{-1}\left(\frac{1}{\sqrt{5}}\right)+\cos^{-1}(x)=\frac{\pi}{4}$

A6 Let $\theta=\sin^{-1}\frac{1}{\sqrt{5}}\Rightarrow\sin\theta=\frac{1}{\sqrt{5}}$



$\tan\theta=\frac{1}{2}\Rightarrow\theta=\tan^{-1}\frac{1}{2}$

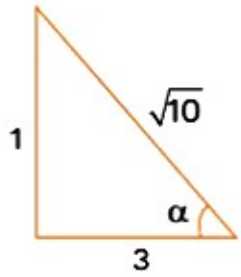
So, $\sin^{-1}\left(\frac{1}{\sqrt{5}}\right)+\cos^{-1}(x)=\frac{\pi}{4}$

$\Rightarrow\tan^{-1}\frac{1}{2}+\cos^{-1}(x)=\tan^{-1}(1)$

$\Rightarrow\cos^{-1}(x)=\tan^{-1}(1)-\tan^{-1}\left(\frac{1}{2}\right)$

$=\tan^{-1}\left(\frac{1-\frac{1}{2}}{1+\frac{1}{2}}\right)$

$=\tan^{-1}\left(\frac{1}{3}\right)$



$\alpha=\tan^{-1}\left(\frac{1}{3}\right)=\cos^{-1}\left(\frac{3}{\sqrt{10}}\right)$

$\Rightarrow\cos^{-1}x=\cos^{-1}\frac{3}{\sqrt{10}}$

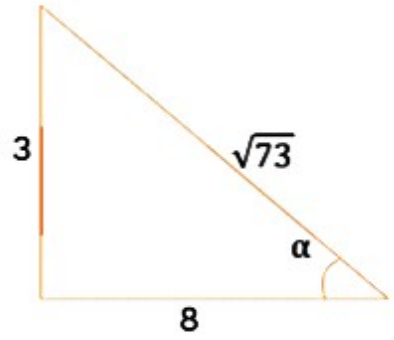
$\Rightarrow x=\frac{3}{\sqrt{10}}$

Q7 $\sin^{-1}\left(\frac{3}{\sqrt{73}}\right)+\cos^{-1}\left(\frac{11}{\sqrt{146}}\right)+\cot^{-1}\left(\sqrt{3}\right)$ **equals**

A7 Let $\alpha=\sin^{-1}\left(\frac{3}{\sqrt{73}}\right)$

$\alpha=\tan^{-1}\left(\frac{3}{8}\right)$

Let $\beta = \cos^{-1}\left(\frac{11}{\sqrt{146}}\right)$

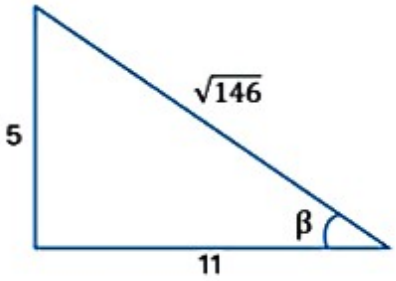


$\cos\beta = \frac{11}{\sqrt{146}}$

$\beta = \tan^{-1}\left(\frac{5}{11}\right)$

$\sin^{-1}\left(\frac{3}{\sqrt{73}}\right) + \cos^{-1}\left(\frac{11}{\sqrt{146}}\right) + \cot^{-1}\left(\sqrt{3}\right)$

$= \tan^{-1}\left(\frac{3}{8}\right) + \tan^{-1}\left(\frac{5}{11}\right) + \frac{\pi}{6}$



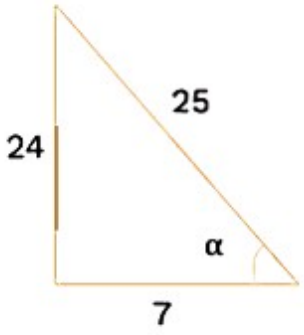
$= \tan^{-1}\left(\frac{\frac{3}{8}+\frac{5}{11}}{1-\frac{3}{8}\times\frac{5}{11}}\right) + \frac{\pi}{6}$

$= \tan^{-1}(1) + \frac{\pi}{6}$

$= \frac{\pi}{4} + \frac{\pi}{6} = \frac{5\pi}{12}$

Q8 **Which is greater** $\cos^{-1}\left(\frac{7}{25}\right) + \cos^{-1}\left(\frac{3}{5}\right)$ or $\cot^{-1}(-1)$

A8 Let $\alpha = \cos^{-1}\left(\frac{7}{25}\right)$



$\alpha = \tan^{-1}\left(\frac{24}{7}\right)$

Let $\beta = \cos^{-1}\left(\frac{3}{5}\right)$

$\cos\beta = \frac{3}{5}$

$\beta = \tan^{-1}\frac{4}{3}$

$\cos^{-1}\frac{7}{25} + \cos^{-1}\frac{3}{5} = \tan^{-1}\left(\frac{24}{7}\right) + \tan^{-1}\left(\frac{4}{3}\right)$

$$= \pi + \tan^{-1}\left(\frac{\frac{24}{7}+\frac{4}{3}}{1-\frac{24}{7}\times\frac{4}{3}}\right)\left(\text{as } \frac{24}{7}\times\frac{4}{3} > 1\right)$$

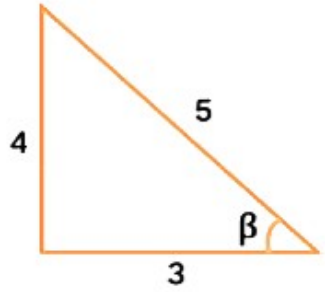
$$= \pi + \tan^{-1}\left(\frac{100}{-75}\right)$$

$$= \pi - \tan^{-1}\left(\frac{4}{3}\right)$$

$= \pi -$ (greater than $\pi/4$).

$$\Rightarrow \cos^{-1}\frac{7}{25} + \cos^{-1}\frac{3}{5} < \frac{3\pi}{4} = \cot^{-1}(-1)$$

So, $\cot^{-1}(-1)$ is greater



Q9 **Find** $A : 2\cos^{-1}\left(\frac{3}{\sqrt{13}}\right) = \tan^{-1}A$

A9 $2\theta = \tan^{-1}A \quad \left(\text{Let } \theta = \cos^{-1}\frac{3}{\sqrt{13}}\right)$

$$\Rightarrow \tan 2\theta = A \qquad \Rightarrow \frac{2\tan\theta}{1-\tan^2\theta} = A$$

$$\Rightarrow \frac{2\times\frac{2}{3}}{1-\left(\frac{2}{3}\right)^2} = A \qquad \Rightarrow A = \frac{12}{5}$$

Q10 **Find B if** $\frac{1}{2}\cos^{-1}\frac{7}{25} = \tan^{-1}B$

A10 $\frac{1}{2}\theta = \tan^{-1}B \quad \left(\text{Let } \theta = \cos^{-1}\frac{7}{25}\right)$

$$\Rightarrow \tan\frac{\theta}{2} = B$$

$$\Rightarrow B = \frac{1-\cos\theta}{\sin\theta}$$

$$\Rightarrow B = \frac{1-\frac{7}{25}}{\frac{24}{25}} \qquad \Rightarrow B = \frac{3}{4}$$

Property-VI :

(a) $\sin^{-1}x + \sin^{-1}y = \begin{cases} \sin^{-1}\left(x\sqrt{1-y^2} + y\sqrt{1-x^2}\right) & \text{if } x, y \geq 0, x^2 + y^2 \leq 1 \\ \pi - \sin^{-1}\left(x\sqrt{1-y^2} + y\sqrt{1-x^2}\right) & \text{if } x, y \geq 0, x^2 + y^2 > 1 \end{cases}$

Proof :

Let $x = \sin A, \quad y = \sin B, \quad x, y \geq 0$

$\sin(A + B) = \sin A \cos B + \cos A \sin B$

$= x\sqrt{1-y^2} + y\sqrt{1-x^2}$

$\sin^{-1}(\sin(A+B)) = \sin^{-1}\left(x\sqrt{1-y^2} + y\sqrt{1-x^2}\right)$ ...(i)

Also $A, B \in \left[0, \frac{\pi}{2}\right]$ $\quad A + B \in [0, \pi]$

Make two case

Case-I :

$A + B \in \left[0, \frac{\pi}{2}\right] \Rightarrow A + B \leq \frac{\pi}{2} \Rightarrow A \leq \frac{\pi}{2} - B \Rightarrow \sin^{-1}x \leq \frac{\pi}{2} - \sin^{-1}y$

$\Rightarrow \sin(\sin^{-1}x) \leq \sin\left(\frac{\pi}{2} - \sin^{-1}y\right)$

$\Rightarrow x \leq \cos(\sin^{-1}y)$

$\Rightarrow x \leq \sqrt{1-y^2}$

$\Rightarrow x^2 \leq 1 - y^2$

$\Rightarrow x^2 + y^2 \leq 1$

By (i) $A + B = \sin^{-1}\left(x\sqrt{1-y^2} + y\sqrt{1-x^2}\right)$

$\sin^{-1}x + \sin^{-1}y = \sin^{-1}\left(x\sqrt{1-y^2} + y\sqrt{1-x^2}\right)$

Case-II :

$$A+B\in\left(\frac{\pi}{2},\pi\right]\Rightarrow A+B>\frac{\pi}{2}\Rightarrow A>\frac{\pi}{2}-B\Rightarrow \sin^{-1}x>\frac{\pi}{2}-\sin^{-1}y$$

$$\Rightarrow \sin\left(\sin^{-1}x\right)>\sin\left(\frac{\pi}{2}-\sin^{-1}y\right)$$

$$\Rightarrow x>\cos\left(\sin^{-1}y\right)$$

$$\Rightarrow x>\sqrt{1-y^2}$$

$$\Rightarrow x^2+y^2>1$$

By (i) $\pi-(A+B)=\sin^{-1}\left(x\sqrt{1-y^2}+y\sqrt{1-x^2}\right)$

$$\Rightarrow \sin^{-1}x+\sin^{-1}y=\pi-\sin^{-1}\left(x\sqrt{1-y^2}+y\sqrt{1-x^2}\right)$$

Q1 **Find whether** $\sin^{-1}\frac{3}{5}+\sin^{-1}\frac{12}{13}$ **is acute/obtuse**

A1 **Method-I** $\left(\frac{3}{5}\right)^2+\left(\frac{12}{13}\right)^2>1\Rightarrow$ obtuse

Method-II $\tan^{-1}\frac{3}{4}+\tan^{-1}\frac{12}{5}$

$$xy=\frac{3}{4}\times\frac{12}{5}>1\Rightarrow \text{obtuse}$$

Point to Remember!!!

Generally, conversion in $\tan^{-1}$ is considered as better.

Property-VI :

(b) $\sin^{-1}x-\sin^{-1}y=\sin^{-1}\left[x\sqrt{1-y^2}-y\sqrt{1-x^2}\right]$ where $x\in[0,1], y\in[0,1]$

$\cos^{-1}x+\cos^{-1}y=\cos^{-1}\left[xy-\sqrt{1-x^2}\ \sqrt{1-y^2}\right]$ where $x\in[0,1], y\in[0,1]$

$\cos^{-1}x-\cos^{-1}y=\cos^{-1}\left[xy+\sqrt{1-x^2}\ \sqrt{1-y^2}\right]$ where $0\le x<y\le 1$

Q1 If $\cos^{-1}x + \cos^{-1}y + \cos^{-1}z = \pi$ then prove that $x^2 + y^2 + z^2 + 2xyz = 1$

A1 $\cos^{-1}x + \cos^{-1}y = \pi - \cos^{-1}z$

Take cosine on both sides

$\cos\left(\cos^{-1}x + \cos^{-1}y\right) = \cos\left(\pi - \cos^{-1}z\right)$

$\Rightarrow \cos\left(\cos^{-1}x\right)\cos\left(\cos^{-1}y\right) - \sin\left(\cos^{-1}x\right)\sin\left(\cos^{-1}y\right) = -\cos\left(\cos^{-1}z\right)$

$\Rightarrow xy - \sqrt{1-x^2}\sqrt{1-y^2} = -z$

$\Rightarrow xy + z = \sqrt{(1-x^2)(1-y^2)}$

Squaring both sides

$\Rightarrow (xy + z)^2 = (1-x^2)(1-y^2)$

$\Rightarrow x^2y^2 + 2xyz + z^2 = 1 - x^2 - y^2 + x^2y^2$

$\Rightarrow x^2 + y^2 + z^2 + 2xyz = 1$

Q2 **Show that** $\cos^{-1}\sqrt{\frac{2}{3}} - \cos^{-1}\frac{\sqrt{6}+1}{2\sqrt{3}} = \frac{\pi}{6}$

A2 **Method I :**

$$\text{LHS} = \cos^{-1}\left(\sqrt{\frac{2}{3}}\frac{\sqrt{6}+1}{2\sqrt{3}} + \sqrt{1-\frac{2}{3}}\sqrt{1-\left(\frac{6+1+2\sqrt{6}}{12}\right)}\right)$$

$$= \cos^{-1}\left(\frac{\sqrt{6}+1}{3\sqrt{2}} + \frac{1}{\sqrt{3}}\sqrt{\frac{5-2\sqrt{6}}{12}}\right)$$

Let $\sqrt{5-2\sqrt{6}} = a - b$

Squaring $\sqrt{5-2\sqrt{6}} = a^2 + b^2 - 2ab$

By hit and trial $a = \sqrt{3}$, $b = \sqrt{2}$

$$\text{So, LHS} = \cos^{-1}\left(\frac{\sqrt{6}+1}{3\sqrt{2}} + \frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}\times 2\sqrt{3}}\right)$$

$$= \cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{6}$$

Method II :

$$\text{LHS} = \tan^{-1}\frac{1}{\sqrt{2}} - \tan^{-1}\left(\frac{\sqrt{3}-\sqrt{2}}{1+\sqrt{3}\cdot\sqrt{2}}\right) = \tan^{-1}\sqrt{\frac{1}{2}} - \left(\tan^{-1}\sqrt{3} - \tan^{-1}\sqrt{2}\right)$$

$$= \cot^{-1}\sqrt{2} - \tan^{-1}\sqrt{3} + \tan^{-1}\sqrt{2} = \frac{\pi}{2} - \tan^{-1}\sqrt{3} = \frac{\pi}{6}$$

Property–VII :

$$\tan^{-1}x + \tan^{-1}y + \tan^{-1}z = \begin{cases} \tan^{-1}\left[\dfrac{x+y+z-xyz}{1-(xy+yz+zx)}\right] & \text{if } x,y,z>0,\ xy+yz+zx<1 \\ \pi + \tan^{-1}\left[\dfrac{x+y+z-xyz}{1-(xy+yz+zx)}\right] & \text{if } x,y,z>0,\ xy+yz+zx>1 \end{cases}$$

Proof : Case-I $xy < 1$

$$\tan^{-1}x + \tan^{-1}y + \tan^{-1}z = \tan^{-1}\frac{x+y}{1-xy} + \tan^{-1}z$$

$$= \tan^{-1}\frac{\left(\frac{x+y}{1-xy}+z\right)}{1-\left(\frac{x+y}{1-xy}\right)z} \qquad \text{if } \frac{x+y}{1-xy}z < 1$$

$$= \tan^{-1}\frac{x+y+z-xyz}{1-(xy+yz+zx)} \qquad \text{if } xy+yz+zx<1 \text{ and } xy<1$$

$$= \pi + \tan^{-1}\frac{x+y+z-xyz}{1-(xy+yz+zx)} \qquad \text{if } xy+yz+zx>1 \text{ and } xy<1$$

Similarly, make case-II $xy > 1$

Q1 **Let r, s, t be roots of the equation x(x – 2) (3x – 7) = 2. Prove that its roots r, s and t are real and positive. Compute $\tan^{-1}r + \tan^{-1}s + \tan^{-1}t$.**

A1 r, s, t are point of intersection of graph
$y = x(x-2)(3x-7)$ and $y = 2$

$r \in (0, 1),\ s \in (1, 2),\ t \in \left(\frac{7}{3}, 3\right)$

$x\left(3x^2 - 13x + 14\right) = 2$

$\Rightarrow 3x^3 - 13x^2 + 14x - 2 = 0$

$$\Rightarrow r + s + t = \frac{13}{3} \quad \ldots(i)$$

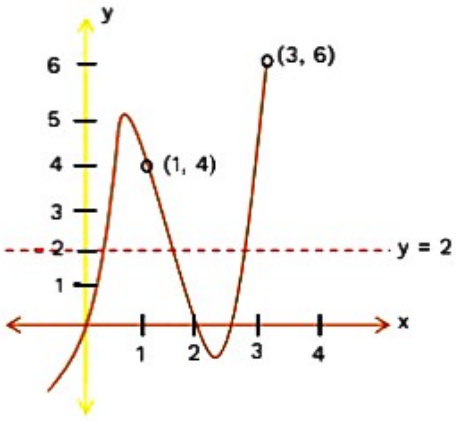
$$\Rightarrow rs + st + tr = \frac{14}{3} \quad \ldots(ii)$$

$$\Rightarrow rst = \frac{2}{3}. \quad \ldots(iii)$$

$$\tan^{-1}r + \tan^{-1}s + \tan^{-1}t = \pi + \tan^{-1}\left(\frac{r + s + t - rst}{1 - (rs + st + tr)}\right)$$

$$= \pi + \tan^{-1}\left(\frac{\frac{13}{3} - \frac{2}{3}}{1 - \frac{14}{3}}\right)$$

$$= \pi + \tan^{-1}(-1) = \frac{3\pi}{4}$$



Q2 **Point (x, y) satisfying the equation** $\sin^{-1}x + \cos^{-1}y + \cos^{-1}(2xy) = \frac{\pi}{2}$ **lies on**

(A) the bisector of the first and third quadrant
(B) bisector of the second and fourth quadrant
(C) the rectangle formed by the lines x = ± 1 and y = ± 1
(D) a unit circle with centre at the origin

A2 **(D)**

$\cos^{-1}y + \cos^{-1}(2xy) = \frac{\pi}{2} - \sin^{-1}x = \cos^{-1}x$

Now, take cos on both sid

$\Rightarrow \cos\left(\cos^{-1}(2xy)\right) = \cos\left(\cos^{-1}x - \cos^{-1}y\right)$

$\Rightarrow 2xy = xy + \sqrt{1-x^2}\sqrt{1-y^2}$

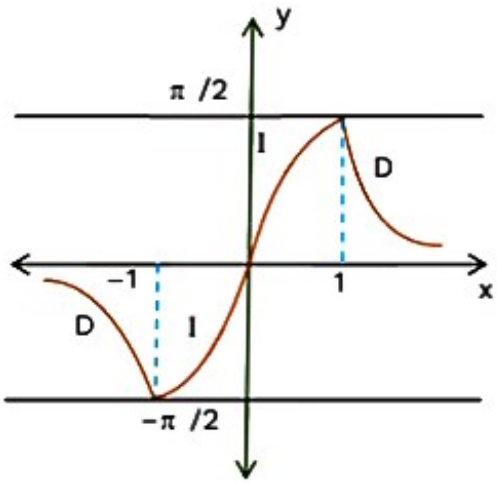
$\Rightarrow xy = \sqrt{1-x^2}\sqrt{1-y^2}$

Squaring on both sides

$\Rightarrow (xy)^2 = (1-x^2)(1-y^2)$

$\Rightarrow x^2 + y^2 = 1$

Simplification & Transformation of Inverse functions by elementry substitution and their graphs:



(1) $\sin^{-1}\dfrac{2x}{1+x^2} = \begin{cases} 2\tan^{-1}x & \text{if } -1 \le x \le 1 \\ \pi - 2\tan^{-1}x & \text{if } x \ge 1 \\ -\pi - 2\tan^{-1}x & \text{if } x \le -1 \end{cases}$

Domain :

$$-1 \le \frac{2x}{1+x^2} \le 1$$

$\Rightarrow \quad -1 - x^2 \le 2x \le 1 + x^2$

$\Rightarrow \quad -1 - x^2 \le 2x \cap 2x \le 1 + x^2$

$\Rightarrow \quad x^2 + 2x + 1 \ge 0 \cap x^2 - 2x + 1 \ge 0$

$\Rightarrow \quad (x+1)^2 \ge 0 \cap (x-1)^2 \ge 0$

$\Rightarrow \quad x \in R \cap x \in R \Rightarrow x \in R$

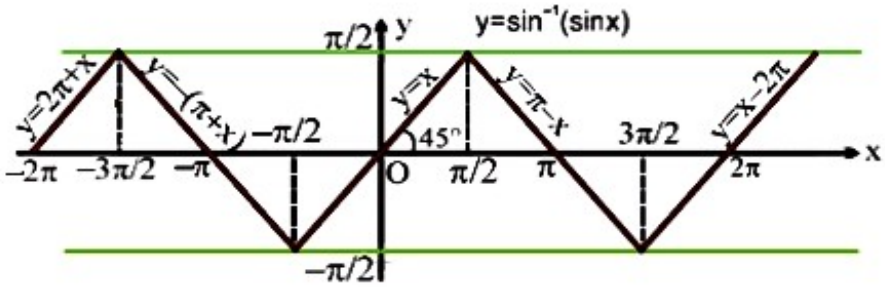
Point :

Expression	Domain	Assumption
$\sqrt{1-x^2}$	$[-1, 1]$	$x = \sin\theta,\ \cos\theta$
$\sqrt{x^2-1}$	$(-\infty,-1] \cup [1,\infty)$	$x = \sec\theta,\ \operatorname{cosec}\theta$
$\sqrt{x^2+1}$	$(-\infty,\infty)$	$x = \tan\theta,\ \cot\theta$

$$f(x) = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$$

$$f(-x) = \sin^{-1}\left(\frac{-2x}{1+x^2}\right)$$

$$= -\sin^{-1}\left(\frac{2x}{1+x^2}\right) = -f(x)$$



$\Rightarrow$ f(x) is odd function.

Let $x = \tan\theta$, where $\theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

$$f(x) = \sin^{-1}\left(\frac{2\tan\theta}{1+\tan^2\theta}\right)$$

$$= \sin^{-1}(\sin 2\theta)$$

Case-I :

$$-\frac{\pi}{2} \le 2\theta \le \frac{\pi}{2} \Rightarrow -\frac{\pi}{4} \le \theta \le \frac{\pi}{4} \Rightarrow \tan\left(-\frac{\pi}{4}\right) \le \tan\theta \le \tan\frac{\pi}{4} \Rightarrow -1 \le x \le 1$$

$$f(x) = \sin^{-1}(\sin 2\theta) = 2\theta = 2\tan^{-1} x$$

Case-II :

$$\frac{\pi}{2} < 2\theta < \pi \Rightarrow \frac{\pi}{4} < \theta < \frac{\pi}{2} \Rightarrow \tan\frac{\pi}{4} < \tan\theta < \tan\frac{\pi}{2} \Rightarrow 1 < x < \infty$$

$$f(x) = \sin^{-1}(\sin 2\theta)$$

$$= \pi - 2\theta$$

$$= \pi - 2\tan^{-1} x$$

Case-III :

$$-\pi < 2\theta < -\frac{\pi}{2} \Rightarrow -\frac{\pi}{2} < \theta < -\frac{\pi}{4} \Rightarrow -\infty < \tan\theta < \tan\left(-\frac{\pi}{4}\right) \Rightarrow -\infty < x < -1$$

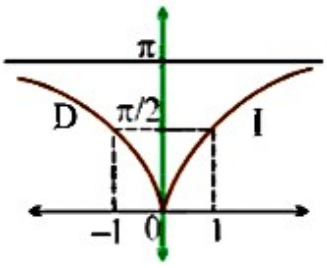
$$f(x) = \sin^{-1}(\sin 2\theta)$$

$$= -\pi - 2\theta = -\pi - 2\tan^{-1} x$$

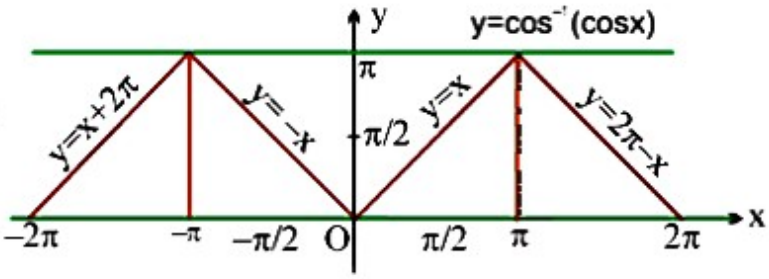
(2) $\cos^{-1}\frac{1-x^2}{1+x^2} = \begin{cases} 2\tan^{-1}x & x \ge 0 \\ -2\tan^{-1}x & x < 0 \end{cases}$

Properties :

(i) f(x) is even function
(ii) Domain is R



Proof :



Let $x = \tan\theta,\ \theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \Rightarrow \theta = \tan^{-1} x$

$$f(x) = \cos^{-1}\left(\frac{1-\tan^2\theta}{1+\tan^2\theta}\right)$$

$$= \cos^{-1}(\cos 2\theta)$$

Now, use graph of $y = \cos^{-1}(\cos x)$

$$f(x) = \begin{cases} 2\theta = 2\tan^{-1}x & \text{when } 2\theta \in [0,\pi) \Rightarrow \theta \in \left[0, \frac{\pi}{2}\right) \Rightarrow \tan\theta = x \in [0,\infty) \\ -2\theta = -2\tan^{-1}x & \text{when } 2\theta \in (-\pi, 0) \Rightarrow \theta \in \left(-\frac{\pi}{2}, 0\right) \Rightarrow \tan\theta = x \in (-\infty, 0) \end{cases}$$

Q1 Solve for x : $3\cos^{-1}x = \sin^{-1}\left(\sqrt{1-x^2}\,(4x^2-1)\right)$

A1 Let $x = \cos\theta,\ \theta \in [0,\pi]$

$$3\cos^{-1}(\cos\theta) = \sin^{-1}\left(\sqrt{1-\cos^2\theta}\,(4\cos^2\theta - 1)\right)$$

$$\Rightarrow 3\theta = \sin^{-1}\left(\sin\theta(4 - 4\sin^2\theta - 1)\right)$$

$$= \sin^{-1}(3\sin\theta - 4\sin^3\theta)$$

$$\Rightarrow 3\theta = \sin^{-1}(\sin 3\theta)$$

$\sin^{-1}(\sin 3\theta) = 3\theta$, when $3\theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ but here $\theta \in [0,\ \pi]$

$$\Rightarrow 3\theta \in \left[0, \frac{\pi}{2}\right]$$

$$\Rightarrow \theta \in \left[0, \frac{\pi}{6}\right]$$

$$\Rightarrow \cos\theta \in \left[\frac{\sqrt{3}}{2}, 1\right]$$

$$\Rightarrow x \in \left[\frac{\sqrt{3}}{2}, 1\right]$$

Q2 **Prove that** $\tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)=\frac{1}{2}\tan^{-1}x, \forall x \in R-\{0\}$

A2 Let $x=\tan\theta,\ \theta\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$

$$\text{LHS} = \tan^{-1}\left(\frac{\sec\theta-1}{\tan\theta}\right)$$

$$= \tan^{-1}\left(\frac{1-\cos\theta}{\sin\theta}\right)$$

$$= \tan^{-1}\left(\frac{2\sin^2\frac{\theta}{2}}{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}\right)$$

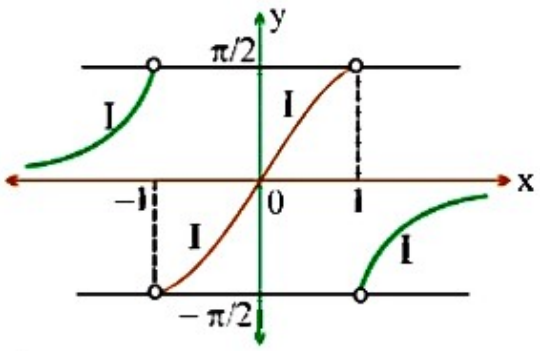
$$= \tan^{-1}\left(\tan\frac{\theta}{2}\right)$$

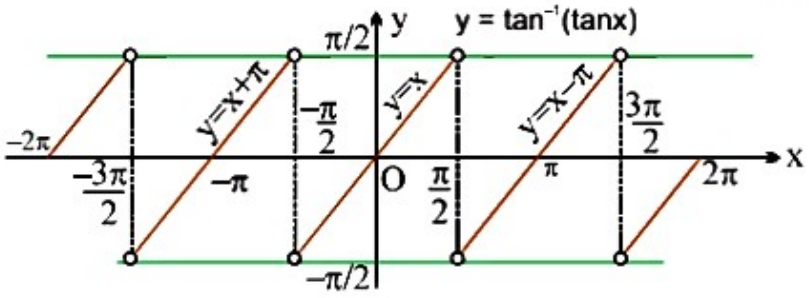
$$= \frac{\theta}{2}\left(\text{as } \theta\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)\right)$$

$$= \frac{\tan^{-1}x}{2} = \text{RHS}$$

Hence proved.

(3) $\tan^{-1}\frac{2x}{1-x^2}=\begin{cases}\pi+2\tan^{-1}x & x<-1\\ 2\tan^{-1}x & -1<x<1\\ 2\tan^{-1}x-\pi & x>1\end{cases}$





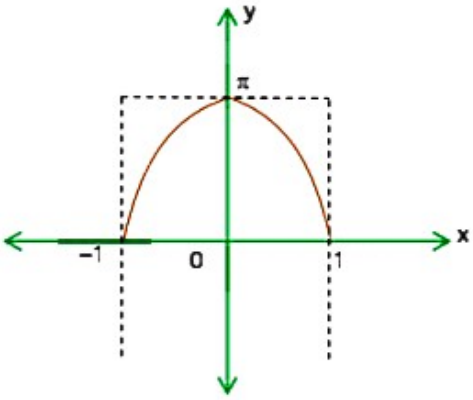
Q3 **Let** $f(x) = \sin^{-1}\frac{2x}{1+x^2} + 2\tan^{-1}x$ **. Find cos(f(10))**

A3 $f(x) = \pi - 2\tan^{-1}x + 2\tan^{-1}x, \quad \text{for } x > 1$

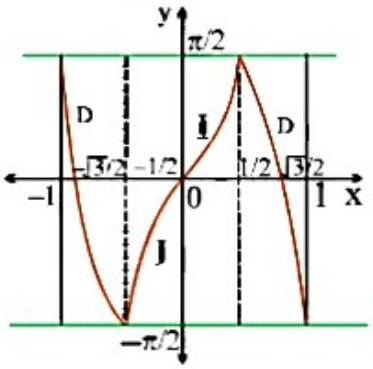
$= \pi$

$\cos(f(10)) = \cos\pi = -1$

(4) $\cos^{-1}\left(2x^2 - 1\right) = \begin{cases} 2\cos^{-1}x & \text{if } x \in [0,1] \\ 2\pi - 2\cos^{-1}x & \text{if } x \in [-1,0] \end{cases}$



(5) $\sin^{-1}\left(3x - 4x^3\right) = \begin{cases} -\left(\pi + 3\sin^{-1}x\right) & \text{if } -1 \le x \le -1/2 \\ 3\sin^{-1}x & \text{if } -1/2 \le x \le 1/2 \\ \pi - 3\sin^{-1}x & \text{if } 1/2 \le x \le 1 \end{cases}$



Proof :

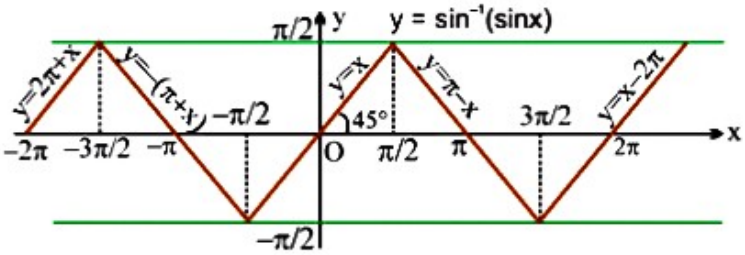
Let $x = \sin\theta, \ \theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

$y = \sin^{-1}\left(3\sin\theta - 4\sin^3\theta\right)$

$= \sin^{-1}\left(\sin 3\theta\right)$

Here, $3\theta \in \left[-\frac{3\pi}{2}, \frac{3\pi}{2}\right]$

Use graph of $y = \sin^{-1}(\sin x)$



Case-I :

$3\theta \in \left[-\frac{3\pi}{2}, -\frac{\pi}{2}\right] \Rightarrow \theta \in \left[-\frac{\pi}{2}, -\frac{\pi}{6}\right] \Rightarrow x = \sin\theta \in \left[-1, -\frac{1}{2}\right]$

$y = \sin^{-1}\left(\sin 3\theta\right)$

$= -\left(\pi + 3\theta\right)$

$= -\left(\pi + 3\sin^{-1}x\right)$

Case-II :

$3\theta \in \left[-\frac{\pi}{2},\frac{\pi}{2}\right] \Rightarrow \theta \in \left[-\frac{\pi}{6},\frac{\pi}{6}\right] \Rightarrow x = \sin\theta \in \left[-\frac{1}{2},-\frac{1}{2}\right]$

$y = \sin^{-1}(\sin 3\theta)$

$= 3\theta$

$= 3\sin^{-1}x$

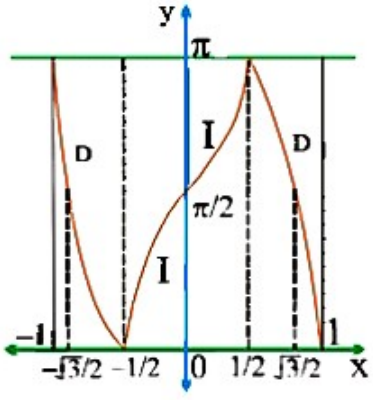
Case-III :

$3\theta \in \left[\frac{\pi}{2},\frac{3\pi}{2}\right] \Rightarrow \theta \in \left[\frac{\pi}{6},\frac{\pi}{2}\right] \Rightarrow x = \sin\theta \in \left[\frac{1}{2},1\right]$

$y = \sin^{-1}(\sin 3\theta)$

$= \pi - 3\theta$

$= \pi - 3\sin^{-1}x$



(6) $\cos^{-1}(4x^3 - 3x) = \begin{cases} 3\cos^{-1}x - 2\pi & \text{if} \quad -1 \le x \le -1/2 \\ 2\pi - 3\cos^{-1}x & \text{if} \quad -1/2 \le x \le 1/2 \\ 3\cos^{-1}x & \text{if} \quad 1/2 \le x \le 1 \end{cases}$

Proof :

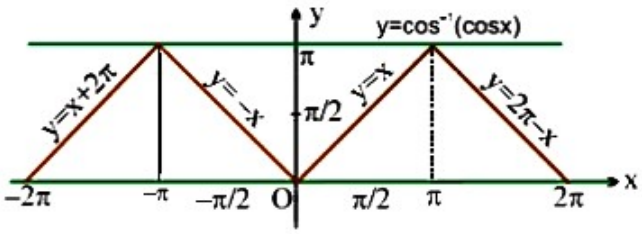
Let $x = \cos\theta,\ \theta \in [0,\pi]$

$y = \cos^{-1}(4\cos^3\theta - 3\cos\theta)$

$= \cos^{-1}(\cos 3\theta)$

Here $3\theta \in [0,3\pi]$

Now, using graph of $y = \cos^{-1}(\cos x)$



Case-I :

$3\theta \in [0,\pi] \Rightarrow \theta \in \left[0,\frac{\pi}{3}\right] \Rightarrow \cos\theta = x \in \left[\frac{1}{2},1\right]$

$y = \cos^{-1}(\cos 3\theta)$

$= 3\theta$

$= 3\cos^{-1}x$

Case-II :

$$3\theta \in [\pi, 2\pi] \Rightarrow \theta \in \left[\frac{\pi}{3}, \frac{2\pi}{3}\right] \Rightarrow x = \cos\theta \in \left[-\frac{1}{2}, \frac{1}{2}\right]$$

$$y = \cos^{-1}(\cos 3\theta)$$

$$= 2\pi - 3\theta$$

$$= 2\pi - 3\cos^{-1}x$$

Case-III :

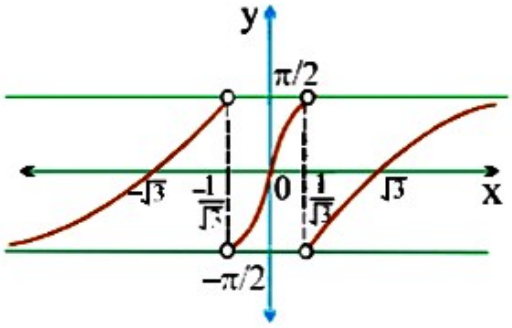
$$3\theta \in [2\pi, 3\pi] \Rightarrow \theta \in \left[\frac{2\pi}{3}, \pi\right] \Rightarrow x = \cos\theta \in \left[-1, -\frac{1}{2}\right]$$

$$y = \cos^{-1}(\cos 3\theta)$$

$$= 3\theta - 2\pi$$

$$= 3\cos^{-1}x - 2\pi$$

(7) $$\tan^{-1}\frac{3x - x^3}{1 - 3x^2} = \begin{cases} 3\tan^{-1}x & \text{if } -\frac{1}{\sqrt{3}} < x < \frac{1}{\sqrt{3}} \\ -\pi + 3\tan^{-1}x & \text{if } x > \frac{1}{\sqrt{3}} \\ \pi + 3\tan^{-1}x & \text{if } x < -\frac{1}{\sqrt{3}} \end{cases}$$



Proof :

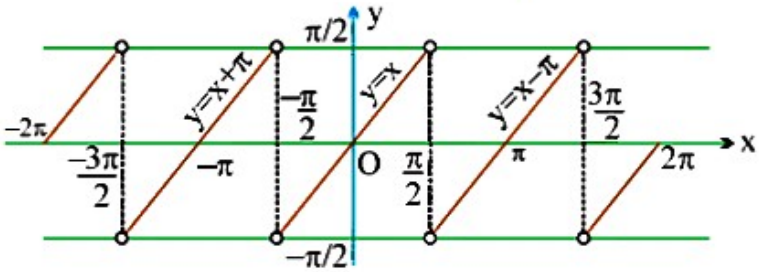
Let $x = \tan\theta,\ \theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

$$y = \tan^{-1}\left(\frac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta}\right)$$

$$= \tan^{-1}(\tan 3\theta)$$

Here, $3\theta \in \left[-\frac{3\pi}{2}, \frac{3\pi}{2}\right]$

Now, using graph of $y = \tan^{-1}(\tan x)$



Case-I :

$$3\theta \in \left(-\frac{3\pi}{2}, -\frac{\pi}{2}\right) \Rightarrow \theta \in \left(-\frac{\pi}{2}, -\frac{\pi}{6}\right) \Rightarrow \tan\theta = x \in \left(-\infty, -\frac{1}{\sqrt{3}}\right)$$

$$y = \tan^{-1}(\tan 3\theta)$$

$$= \pi + 3\theta$$

$$= \pi + 3\tan^{-1}x$$

Case-II :

$$3\theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \Rightarrow \theta \in \left(-\frac{\pi}{6}, \frac{\pi}{6}\right) \Rightarrow \tan\theta = x \in \left(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$$

$$y = \tan^{-1}(\tan 3\theta)$$

$$= 3\theta$$

$$= 3\tan^{-1}x$$

Case-III :

$$3\theta \in \left(\frac{\pi}{2}, \frac{3\pi}{2}\right) \Rightarrow \theta \in \left(\frac{\pi}{6}, \frac{\pi}{2}\right) \Rightarrow \tan\theta = x \in \left(\frac{1}{\sqrt{3}}, \infty\right)$$

$$y = \tan^{-1}(\tan 3\theta)$$

$$= 3\theta - \pi$$

$$= 3\tan^{-1}x - \pi$$

Q4 **Show that** $\tan^{-1}x = 2\tan^{-1}\left[\operatorname{cosec}\left(\tan^{-1}x\right) - \tan\left(\cot^{-1}x\right)\right]$

A4 Let $x = \tan\theta$, $\theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \Rightarrow \tan^{-1}x = \theta$

$$\text{RHS} = 2\tan^{-1}\left[\operatorname{cosec}\theta - \tan\left(\frac{\pi}{2} - \theta\right)\right]$$

$$= 2\tan^{-1}\left[\frac{1}{\sin\theta} - \cot\theta\right]$$

$$= 2\tan^{-1}\left[\frac{1-\cos\theta}{\sin\theta}\right]$$

$$= 2\tan^{-1}\left(\frac{2\sin^2\frac{\theta}{2}}{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}\right)$$

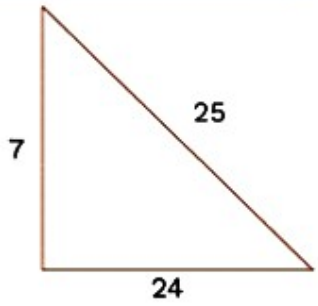
$$= 2\tan^{-1}\left(\tan\frac{\theta}{2}\right)$$

$$= 2\times\frac{\theta}{2} = \theta$$

$= \tan^{-1}(x)$ = LHS **Hence proved.**

Q5 **Find x;** $2\cot^{-1}2 - \cos^{-1}\left(\frac{4}{5}\right) = \text{cosec}^{-1}x$.

A5 $2\tan^{-1}\left(\frac{1}{2}\right) - \tan^{-1}\left(\frac{3}{4}\right) = \text{cosec}^{-1}x$



$$\Rightarrow \tan^{-1}\left(\frac{2\times\frac{1}{2}}{1-\left(\frac{1}{2}\right)^2}\right) - \tan^{-1}\left(\frac{3}{4}\right) = \text{cosec}^{-1}x$$

$$\Rightarrow \tan^{-1}\left(\frac{4}{3}\right) - \tan^{-1}\left(\frac{3}{4}\right) = \text{cosec}^{-1}x$$

$$\Rightarrow \tan^{-1}\left(\frac{\frac{4}{3}-\frac{3}{4}}{1+\frac{4}{3}\times\frac{3}{4}}\right) = \text{cosec}^{-1}x$$

$$\Rightarrow \tan^{-1}\left(\frac{7}{24}\right) = \text{cosec}^{-1}x$$

$$\Rightarrow \text{cosec}^{-1}\left(\frac{25}{7}\right) = \text{cosec}^{-1}x$$

$$\Rightarrow x = \frac{25}{7}$$

Q6 **If** $\cos^{-1}x - \sin^{-1}x = \cos^{-1}x\sqrt{3}$ **, then tell x**

A6 $\cos^{-1}x - \sin^{-1}x = \cos^{-1}x\sqrt{3}$ ... (i)

we know $\cos^{-1}x + \sin^{-1}x = \frac{\pi}{2}$ ... (ii)

(i) – (ii)

$$\frac{\pi}{2} - 2\sin^{-1}x = \cos^{-1}\left(x\sqrt{3}\right)$$

Take cos on both sides

$$\Rightarrow \sin\left(2\sin^{-1}x\right) = \cos\left(\cos^{-1}\left(x\sqrt{3}\right)\right)$$

$$\Rightarrow \sin 2\theta = x\sqrt{3} \qquad \text{where } \theta = \sin^{-1}x$$

$$\Rightarrow 2\sin\theta\cos\theta = x\sqrt{3}$$

$$\Rightarrow 2x\cdot\sqrt{1-x^2} = x\sqrt{3}$$

$$\Rightarrow x = 0 \text{ or } 4(1-x^2) = 3$$

$$\Rightarrow x = 0, \pm\frac{1}{2}$$

Each is satisfying given equation

$$x = 0, \frac{1}{2}, -\frac{1}{2}$$ **Ans.**

Q7 **$\sin[2\cos^{-1}\{\cot(2\tan^{-1}x)\}] = 0$**

A7 $2\cos^{-1}\{\cot(2\tan^{-1}x)\} = n\pi,\ n \in I$

$= 0, \pi, 2\pi$ if $n = 0,1,2$ respectively

Hence $\cos^{-1}\{\cot(2\tan^{-1}x)\} = 0$ or $\frac{\pi}{2}$ or π

or $\cot(2\tan^{-1}x) = 1, 0,$ or -1

(i) $\cot\left(2\tan^{-1}x\right) = 1 \Rightarrow 2\tan^{-1}x = m\pi + \frac{\pi}{4},\ m \in I$

$$\Rightarrow \tan^{-1}x = \frac{m\pi}{2} + \frac{\pi}{8}$$

$$\Rightarrow \tan^{-1}x = -\frac{3\pi}{8}, \frac{\pi}{8} \left(\text{as } \tan^{-1}x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)\right)$$

$$\Rightarrow x = \tan\left(-\frac{3\pi}{8}\right), \tan\left(\frac{\pi}{8}\right)$$

$$\Rightarrow x = -\left(\sqrt{2}+1\right), \sqrt{2}-1$$

(ii) $\cot\left(2\tan^{-1}x\right) = 0 \Rightarrow 2\tan^{-1}x = (2m+1)\frac{\pi}{2},\ m \in I$

$$\Rightarrow \tan^{-1}x = (2m+1)\frac{\pi}{4},\quad m \in I$$

$$\Rightarrow \tan^{-1}x = -\frac{\pi}{4}, \frac{\pi}{4} \qquad \left(\text{as } \tan^{-1}x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)\right)$$

$$\Rightarrow x = \tan\left(-\frac{\pi}{4}\right), \tan\left(\frac{\pi}{4}\right)$$

$$\Rightarrow x = -1, 1$$

(iii) $\cot\left(2\tan^{-1}x\right) = -1 \Rightarrow 2\tan^{-1}x = m\pi - \frac{\pi}{4},\ m \in I$

$$\Rightarrow \tan^{-1}x = (4m-1)\frac{\pi}{8}$$

$$\Rightarrow \tan^{-1}x = -\frac{\pi}{8}, \frac{3\pi}{8} \quad \left(\text{as } \tan^{-1}x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)\right)$$

$$\Rightarrow x = \tan\left(-\frac{\pi}{8}\right), \tan\left(\frac{3\pi}{8}\right) \Rightarrow x = -\left(\sqrt{2}-1\right), \sqrt{2}+1$$

$x = \pm 1, \pm\left(1 \pm \sqrt{2}\right)$ **Ans.**

Q8 **Tell numbers of solution(s) and solve** $\sin^{-1}x + \sin^{-1}2x = \frac{\pi}{3}$

A8 Let $f(x) = \sin^{-1}x + \sin^{-1}2x$

Domain of f(x) is $\left[-\frac{1}{2}, \frac{1}{2}\right]$

Here, both $\sin^{-1}x$ and $\sin^{-1}2x$ are increasing function
Hence, f(x) is also increasing function.

Range of f(x) will be $\left[f\left(-\frac{1}{2}\right), f\left(\frac{1}{2}\right)\right]$

Range is $\left[-\frac{2\pi}{3}, \frac{2\pi}{3}\right]$

So, number of solution(s) of equation is 1.
Again, consider equation

$\alpha + \beta = \frac{\pi}{3}$ where $\sin\alpha =$ x and $\sin\beta =$ 2x and $\beta = \frac{\pi}{3} - \alpha$

now $\sin\beta = \sin\left(\frac{\pi}{3} - \alpha\right)$;

hence $\sin\beta = \sin\frac{\pi}{3}\cos\alpha - \cos\frac{\pi}{3}\sin\alpha$

$$2x = \frac{\sqrt{3}}{2}\sqrt{1-x^2} - \frac{1}{2}x \Rightarrow \frac{5x}{2} = \frac{\sqrt{3}}{2}\sqrt{1-x^2} \Rightarrow 25x^2 = 3\left(1-x^2\right)$$

$$\Rightarrow 28x^2 = 3 \quad \Rightarrow x^2 = \frac{3}{28}$$

$$\Rightarrow x = \frac{\sqrt{3}}{2\sqrt{7}} \quad \left(\text{as } -\frac{\sqrt{3}}{2\sqrt{7}} \text{ is not satisfying given equation}\right)$$

Q9 $\tan^{-1}\left(\frac{x+1}{x-1}\right)+\tan^{-1}\left(\frac{x-1}{x}\right)=\tan^{-1}(-7)$

A9 Take tan on both side.

$$\frac{\tan\left(\tan^{-1}\left(\frac{x+1}{x-1}\right)\right)+\tan\left(\tan^{-1}\left(\frac{x-1}{x}\right)\right)}{1-\tan\left(\tan^{-1}\left(\frac{x+1}{x-1}\right)\right)\tan\left(\tan^{-1}\left(\frac{x-1}{x}\right)\right)}=\tan\left(\tan^{-1}(-7)\right)$$

$$\Rightarrow \frac{\frac{x+1}{x-1}+\frac{x-1}{x}}{1-\left(\frac{x+1}{x-1}\right)\left(\frac{x-1}{x}\right)}=-7$$

$$\Rightarrow \frac{\frac{x^2+x+(x-1)^2}{(x-1)x}}{\frac{(x^2-x)-(x^2-1)}{(x-1)x}}=-7$$

$$\Rightarrow \frac{2x^2-x+1}{-x+1}=-7$$

$$\Rightarrow 2x^2-x+1=7x-7$$

$$\Rightarrow 2x^2-8x+8=0$$

$$\Rightarrow 2(x-2)^2=0$$

$$\Rightarrow x=2$$

Cross Check:

$x = 2$

Given equation becomes

$$\tan^{-1}3+\tan^{-1}\left(\frac{1}{2}\right)=\tan^{-1}(-7)$$

$$\Rightarrow\left(0,\frac{\pi}{2}\right)+\left(0,\frac{\pi}{2}\right)=\left(-\frac{\pi}{2},0\right)$$

Which is not possible.

Hence, No solution.

Q10 **Solve $2\cot^{-1}2 + \cos^{-1}(3/5) = \operatorname{cosec}^{-1}x$**

A10 $\text{LHS} = 2\tan^{-1}\frac{1}{2} + \tan^{-1}\frac{4}{3} = \tan^{-1}\frac{2\cdot\frac{1}{2}}{1-\frac{1}{4}} + \tan^{-1}\left(\frac{4}{3}\right)$

$\Rightarrow 2\tan^{-1}\frac{4}{3} = \operatorname{cosec}^{-1}x$

$\Rightarrow$ No solution (As LHS is greater than $\frac{\pi}{2}$)

Q11 **Solve $\cos^{-1}x > \cos^{-1}x^2$**

A11 Domain: $x \in [-1,1] \cap x^2 \in [-1,1] \Rightarrow x \in [-1,1]$

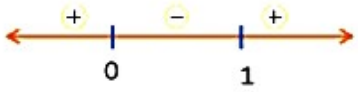
Now, $\cos^{-1}x > \cos^{-1}x^2$

$\Rightarrow x < x^2$ (as $\cos^{-1}x$ is decreasing)

$\Rightarrow x(x-1) > 0$

$\Rightarrow x \in (-\infty, 0) \cup (1, \infty)$

Solution $\cap$ Domain gives $[-1, 0)$



Q12 **Solve $\sin^{-1}x > \cos^{-1}x$**

A12 Domain $x \in [-1, 1]$

Now, $\sin^{-1}x > \cos^{-1}x$

$\Rightarrow \sin^{-1}x > \frac{\pi}{2} - \sin^{-1}x$

$\Rightarrow 2\sin^{-1}x > \frac{\pi}{2}$

$\Rightarrow \sin^{-1}x > \frac{\pi}{4}$

$\Rightarrow x > \frac{1}{\sqrt{2}}$ (as $\sin^{-1}x$ is increasing function)

Solution $\cap$ Domain gives $\left(\frac{1}{\sqrt{2}}, 1\right]$

Q13 **Solve $\sin^{-1}x > \sin^{-1}(1-x)$**

A13 Domain: $1 - x \in [-1, 1] \cap x \in [-1, 1]$

$\Rightarrow x \in [0,2] \cap x \in [-1,1]$

$\Rightarrow x \in [0,1]$

Now, $\sin^{-1}x > \sin^{-1}(1-x)$

$\Rightarrow x > 1 - x \quad \Rightarrow x > \frac{1}{2}$ (as $\sin^{-1}x$ is increasing function)

Solution $\cap$ Domain gives $\left(\frac{1}{2}, 1\right]$

Q14 **Solve $\text{arc}\tan^2 x - 3\,\text{arc}\tan x + 2 > 0$**

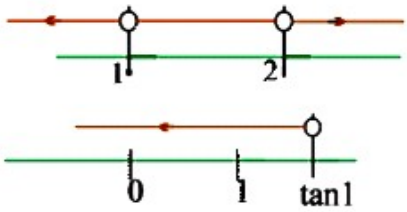
A14 $\left(\tan^{-1}x\right)^2 - 3\tan^{-1}x + 2 > 0$ or $t^2 - 3t + 2 > 0$

$\Rightarrow (t-2)(t-1) > 0 \quad [t = \tan^{-1}(x)]$

$\left(\tan^{-1}x - 2\right)$ is always < 0

$\therefore \quad \tan^{-1}x < 1 \quad \Rightarrow \quad x < \tan 1$

$\therefore \quad x \in (-\infty, \tan 1)$



Q15 **Solve $[\sin^{-1}x] > [\cos^{-1}x]$, where [.] denotes the greatest integer function.**

A15 $[\sin^{-1}x] > [\cos^{-1}x]$

$[\sin^{-1}x]$ can take the values $\{-2, -1, 0, 1\}$ and

$[\cos^{-1}x]$ can take the values $\{0, 1, 2, 3\}$

Hence, $[\sin^{-1}x]$ can be greater than $(\cos^{-1}x)$ only

if $[\sin^{-1}x] = 1$ and $[\cos^{-1}x] = 0$

now $[\sin^{-1}x] = 1 \Rightarrow \quad 1 \le \sin^{-1}x \le \pi/2 \quad (1 \le \sin^{-1}x < 2)$

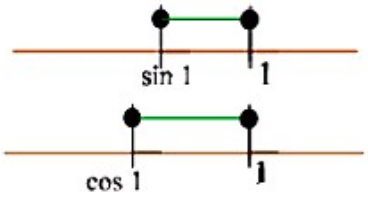
$\Rightarrow \quad \sin 1 \le x \le 1$ ... (i)

and $[\cos^{-1}x] = 0 \Rightarrow 0 \le \cos^{-1}x < 1$

$\Rightarrow \quad \cos 1 < x \le 1$ ... (ii)

By (i) and (ii):

$x \in [\sin 1, 1]$



Summation of Series

Think by method of difference and $\tan^{-1}x - \tan^{-1}y = \tan^{-1}\left(\frac{x-y}{1+xy}\right)$

Q1 $\tan^{-1}\frac{2}{2+1^2+1^4} + \tan^{-1}\frac{4}{2+2^2+2^4} + \tan^{-1}\frac{6}{2+3^2+3^4} + \ldots$

A1 $T_r = \tan^{-1}\frac{2n}{2+n^2+n^4} = \tan^{-1}\frac{2n}{1+(n^2+1)^2-n^2}$

$$=\tan^{-1}\frac{2n}{1+(n^2+1+n)(n^2-n+1)}$$

$$= \tan^{-1}\frac{(n^2+n+1)-(n^2-n+1)}{1+(n^2+n+1)(n^2-n+1)}$$

$T_r = \tan^{-1}(n^2+n+1) - \tan^{-1}(n^2-n+1)$

$T_1 = \tan^{-1}3 - \tan^{-1}1$

$T_2 = \tan^{-1}7 - \tan^{-1}3$

$T_3 = \tan^{-1}13 - \tan^{-1}7$

$\vdots$

$T_n = \tan^{-1}(n^2+n+1) - \tan^{-1}(n^2-n+1)$

$S_n = \tan^{-1}(n^2+n+1) - \tan^{-1}(1)$

$= \tan^{-1}(n^2+n+1) - \frac{\pi}{4}$

$S_\infty = \frac{\pi}{2} - \frac{\pi}{4} \Rightarrow S_\infty = \frac{\pi}{4}$

Q2 **Tell sum of n terms of** $\tan^{-1}\frac{x}{1+(1\times2)x^2} + \tan^{-1}\frac{x}{1+(2\times3)x^2} + \ldots + \tan^{-1}\frac{x}{1+n(n+1)x^2}$

A2 General term $= T_n = \tan^{-1}\left(\frac{x}{1+n(n+1)x^2}\right)$

$$= \tan^{-1}\left[\frac{(n+1)x-nx}{1+(nx)((n+1)x)}\right] = \tan^{-1}((n+1)x) - \tan^{-1}(nx)$$

$T_1 = \tan^{-1}(2x) - \tan^{-1}(x)$

$T_2 = \tan^{-1}(3x) - \tan^{-1}(2x)$

$T_3 = \tan^{-1}(4x) - \tan^{-1}(3x)$

$\vdots$

$T_n = \tan^{-1}((n+1)x) - \tan^{-1}(nx)$

$S_n = \tan^{-1}(n+1)x - \tan^{-1}(x)$

$= \tan^{-1}\left[\dfrac{(n+1)x - x}{1+(n+1)x.x}\right]$

$= \tan^{-1}\left(\dfrac{nx}{1+(n+1)x^2}\right)$

Q3 **Tell** $S_n = \sum_{n=1}^{n} \tan^{-1}\left(\dfrac{4n}{n^4 - 2n^2 + 2}\right)$ and also tell S_∞..

A3 $T_n = \tan^{-1}\left(\dfrac{4n}{n^4 - 2n^2 + 1 + 1}\right) = \tan^{-1}\left(\dfrac{(n+1)^2 - (n-1)^2}{1+(n-1)^2(n+1)^2}\right)$

$= \tan^{-1}(n+1)^2 - \tan^{-1}(n-1)^2$

$T_1 = \tan^{-1}4 - \tan^{-1}0$

$T_2 = \tan^{-1}9 - \tan^{-1}1$

$T_3 = \tan^{-1}16 - \tan^{-1}4$

$T_4 = \tan^{-1}25 - \tan^{-1}9$

$\vdots$

$T_{n-1} = \tan^{-1}(n^2) - \tan^{-1}(n-2)^2$

$T_n = \tan^{-1}(n+1)^2 - \tan^{-1}(n-1)^2$

$S_n = \tan^{-1}(n^2) + \tan^{-1}(n+1)^2 - \tan^{-1}1 - \tan^{-1}0$

$S_n = \tan^{-1}(n+1)^2 + \tan^{-1}n^2 - \dfrac{\pi}{4}$

$S_\infty = \dfrac{\pi}{2} + \dfrac{\pi}{2} - \dfrac{\pi}{4}$

$S_\infty = \dfrac{3\pi}{4}$

Q4 **Tell** $S_n = \sum_{i=1}^{\infty} \tan^{-1}\left(\frac{1}{2i^2}\right)$ and also tell S_∞

A4

$$T_i = \tan^{-1}\left(\frac{1}{2i^2}\right)$$

$$= \tan^{-1}\left(\frac{2}{4i^2}\right)$$

$$= \tan^{-1}\left(\frac{(2i+1)-(2i-1)}{1+4i^2-1}\right)$$

$$= \tan^{-1}(2i+1) - \tan^{-1}(2i-1)$$

$$T_1 = \tan^{-1}3 - \tan^{-1}1$$

$$T_2 = \tan^{-1}5 - \tan^{-1}3$$

$$T_3 = \tan^{-1}7 - \tan^{-1}5$$

$$\vdots$$

$$T_n = \tan^{-1}(2n+1) - \tan^{-1}(2n-1)$$

$$S_n = \tan^{-1}(2n+1) - \frac{\pi}{4}$$

$$S_\infty = \frac{\pi}{2} - \frac{\pi}{4} \Rightarrow S_\infty = \frac{\pi}{4}$$

Q5 **Tell** $\text{cosec}^{-1}\sqrt{5} + \text{cosec}^{-1}\sqrt{65} + \text{cosec}^{-1}\sqrt{325} + \ldots\infty$

A5

$$\tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{1}{8}\right) + \tan^{-1}\left(\frac{1}{18}\right)$$

$$T_n = \tan^{-1}\left(\frac{1}{2n^2}\right)$$

$$= \tan^{-1}\left(\frac{2}{4n^2}\right)$$

$$= \tan^{-1}\left(\frac{(2n+1)-(2n-1)}{1+(2n+1)(2n-1)}\right)$$

$$= \tan^{-1}(2n+1) - \tan^{-1}(2n-1)$$

$T_1 = \tan^{-1}3 - \tan^{-1}1$

$T_2 = \tan^{-1}5 - \tan^{-1}3$

$T_3 = \tan^{-1}7 - \tan^{-1}5$

$\vdots$

$T_n = \tan^{-1}(2n+1) - \tan^{-1}(2n-1)$

$S_n = \tan^{-1}(2n+1) - \frac{\pi}{4}$

$S_\infty = \frac{\pi}{2} - \frac{\pi}{4}$

$S_\infty = \frac{\pi}{4}$

Q6 **Tell** $S_n = \sum_{r=1}^{n}\left(\cos^{-1}\left(\frac{1}{n+1}\right) - \cos^{-1}\left(\frac{1}{n}\right)\right)$ and S_∞

A6 $T_n = \cos^{-1}\left(\frac{1}{n+1}\right) - \cos^{-1}\left(\frac{1}{n}\right)$

$T_1 = \cos^{-1}\left(\frac{1}{2}\right) - \cos^{-1}(1)$

$T_2 = \cos^{-1}\left(\frac{1}{3}\right) - \cos^{-1}\left(\frac{1}{2}\right)$

$\vdots$

$T_n = \cos^{-1}\left(\frac{1}{n+1}\right) - \cos^{-1}\left(\frac{1}{n}\right)$

$S_n = \cos^{-1}\left(\frac{1}{n+1}\right) - \cos^{-1}(1)$

$S_\infty = \cos^{-1}(0) - 0$

$S_\infty = \frac{\pi}{2}$